

Acknowledgements

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Introduction

Many commercial transactions are simple: A seller posts a per-unit price at which buyers can buy; a buyer then chooses how much to buy at the posted price. However, when both the seller and the buyer are firms, the contract governing their trade is often more complex. In addition to a per-unit price, the contract may stipulate fixed fees paid by the buyer to the seller (or by the seller to the buyer), constraints on the amount the buyer can or must buy from the seller, limitations on what the buyer can do with the product, and restrictions on the trade between the buyer and other sellers, and on the trade between the seller and other buyers. *Vertical restraints* is often used as a name for any contractual arrangement that comes in addition to a simple per-unit price. This thesis is about vertical restraints.¹

Why do contracts between firms include vertical restraints? In contrast to consumers, who by definition consume the goods they buy, a downstream firm uses the products it buys from upstream firms as inputs in production. Sometimes the production process is trivial and leaves the input more or less unchanged, for example when a retailer merely resells a product to final consumers. Other times a given input is not recognizable at all in the final product. When choosing what to do with a given input (with which technology to transform it, how to combine it with other inputs, to whom to sell the final product and at which prices) a downstream firm often affects the profit of its supplier. The opposite can also be true. The behavior of the upstream firm (how much to invest in product development or advertising, with how many and which downstream firms to trade) may affect the profit of the downstream firm. The behavior of each firm thus has external effects on its trading partner. Both firms may therefore want to use vertical restraints to control the behavior of their trading partner, and it may be in their joint interest to stipulate

¹ A comprehensive overview of types of vertical restraints and their use is found in Tirole (1988) and Rey and Vergé (2008).

vertical restraints that internalizes the relevant externalities.

However, while a given pair of trading partners may find it profitable to include vertical restraints in their contracts, it is not obvious that this also benefits other firms in the industry, consumers, or society as a whole. A central theme in this thesis is how the use of vertical restraints affects not only the creation and division of profit in a vertical chain, but also which products are produced in an industry, and at which prices these products are made available for the final consumers.

The first chapter, which is co-written by Øivind Anti Nilsen and Lars Sørgard, illustrates how *fixed fees* (or other “non-linear” components of the supply contracts between upstream and downstream firms) can affect the welfare effects of a merger between upstream firms. A theoretical model is developed that illustrates that increases in upstream market power following a merger may lead to changes in the fixed fees rather than in the per-unit wholesale price paid by downstream to upstream firms. If the per-unit wholesale price is unchanged following a merger, the downstream firms will have no incentive to change the prices they in turn charge their customers. The theoretical predictions are tested by examining an upstream merger in the Norwegian market for eggs. This merger appears to have resulted in higher total payments from grocery chains to egg producers, without affecting the consumer prices for eggs. This is consistent with the theoretical prediction that an upstream merger affects the fixed fees and not the per-unit price in the contracts between upstream and downstream firms.

The second chapter considers how the use of *exclusionary contracts*, that is, contracts where a buyer commits not to buy from other sellers, affects the distribution of new and valuable products. A theoretical model is developed that illustrates how an upstream firm selling an input to two competing downstream firms can prevent entry of a potential competitor producing a competing version of the input in question, by signing exclusionary contracts with the buyers. Why would a buyer agree to sign an exclusionary contract with a seller, if this precludes it from obtaining a valuable product from a potential entrant? The explanation given in this chapter is the following: When the buyers are competitors, a buyer’s profit gain from buying from the entrant (rather than only from the established seller) is partly the rival buyer’s loss. The existence of these externalities among the buyers can enable the potential entrant to extract more profit from the industry than its contribution, something that would give the buyers and the incumbent a joint incentive to exclude the entrant. In the model, there always exists equilibria where the entrant is

excluded, even though entry would increase total industry profit, and even when the fixed cost of entry is zero. Perhaps surprisingly, the buyers often gain from the use of exclusionary contracts. The end-consumers will however tend lose from the use of exclusionary contracts, since this restricts the availability of the entrant's product.

The thesis ends with two short articles about *quantity rationing tariffs*. The third chapter illustrates that the anti-competitive use of *resale price-maintenance* (RPM) – that is, restrictions on the price at which a downstream firm can resell a product it buys from an upstream firm – in some types of industries can be mimicked by simple, and less legally dubious, quantity rationing contracts which disincentives the downstream firm from buying more than a given quantity. As a consequence, banning RPM may be ineffective in these industries. The fourth and final chapter reexamines a common result in the existing literature: When two competing firms offer contracts to a common agent, they each get a profit equal to their contribution to the industry profit, while the agent gets the remainder of the industry profit. This chapter shows how the ability to offer menus of quantity rationing contracts on a take-it-or-leave-it basis enables the competing firms to capture the entire industry profit, each getting strictly more than its contribution and leaving nothing for the agent.²

A better understanding of vertical restraints is important for competition authorities. For one, competition authorities must decide which vertical restraints to tolerate. Some vertical restraints have been and are still *per se* unlawful in certain jurisdictions, while others are assessed on a case-to-case basis. Of the restraints discussed in this thesis, RPM and exclusionary contracts have been met with skepticism from competition authorities. Better understanding when these restraints can be expected to have anti-competitive effects is therefore important. A second reason why understanding vertical restraints is crucial is illustrated in the second chapter of this thesis. Even in cases where the use of vertical restraints is not the primary issue (*e.g.*, a horizontal merger case), understanding how market outcomes are affected by the use of vertical restraints may be necessary to make well-informed decisions.

² The fourth chapter has been published in *Economics Letters*, 2016, 138: 72-74.

Bibliography

- Rey, P. and T. Vergé (2008). The Economics of Vertical Restraints. In: *Handbook of Antitrust Economics*. Ed. by P. Buccirossi. MIT Press. Chap. 9, pp. 353–390.
- Tirole, J. (1988). *The Theory of Industrial Organization*. MIT press.

Chapter 1

Upstream merger in a successive oligopoly: Who pays the price?

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Abstract

This article applies a successive oligopoly model, with unobservable non-linear tariffs between upstream and downstream firms, to analyze the possible anti-competitive effects of an upstream merger in the Norwegian food sector (specifically, the market for eggs). The theoretical predictions are that an upstream merger may lead to higher average prices paid by downstream firms and at the same time to no changes in the prices paid by consumers. Consistent with the theoretical predictions, we find that the merger had no effect on consumer prices, but led to higher average prices paid by the downstream firms to the merged firm.

¹ We are indebted to the Norwegian Research Council for financial support to the project “Improving Competition Policy” at SNF. Furthermore, we are grateful for comments and suggestions by Luis Bru, Frode Steen, Carl Shapiro, participants at the CLEEN workshop in Bergen in May 2013, the Nordic Econometric Meeting in Bergen in June 2013, the CRESSE conference in Corfu in July 2013, the EARIE conference in Evora in September 2013, the Peder Sather IO workshop at Berkeley in November 2013, and seminar participants at the Center for Macroeconomic Research at the University of Cologne (CMR). Thanks also to Elin Røsnes at the Norwegian Agricultural Authority for help in providing the data.

1.1 Introduction

Merger control has been especially concerned about horizontal mergers. The reason is obvious. Such mergers may lead to less intense rivalry between firms and, in turn, higher consumer prices. It is argued that the closer are the substitutes the firms produce, the larger is the potential anti-competitive effect.² However, an upstream merger in a successive oligopoly may be poorly captured by such an intuitive and simple reasoning. The existence of a non-linear wholesale price between upstream and downstream firms, such as a two-part tariff consisting of a marginal wholesale price and a fixed fee (or some rebate scheme), may affect the price change for final consumers following an upstream merger. The purpose of this study is to investigate the price effects of an upstream merger. According to the theoretical model applied, an upstream merger that leads to increased market power may result only in a profit shift between upstream and downstream firms, and have no effect on the downstream firms' prices (prices to the final consumers). These theoretical predictions are tested on data from an upstream merger in the Norwegian food sector, more particularly in the egg market. In line with the theoretical predictions, we find that this particular upstream merger led to higher average wholesale prices, but did not lead to a change in consumer prices. The downstream firms, and not the final consumers, paid the price for the upstream merger.

Most models of horizontal mergers neglect the role of the vertical structure in an industry.³ This can be an innocuous assumption, if any price increase from an upstream to a downstream firm due to an upstream merger is passed on to consumers. However, even with a simple linear wholesale price between upstream and downstream firms, the pass-through rate can vary a lot depending on – among other things – the demand function.⁴ If non-linear pricing between upstream and downstream firms is introduced, there is an even less clear relationship between the

² One clear statement of this principle appears in the US Horizontal Merger Guidelines from August 2010, where they focus on diversion ratios: “Diversion ratios between products sold by one merging firm and products sold by the other merging firm can be very informative for assessing unilateral price effects, with higher diversion ratios indicating a greater likelihood of such effects” (p. 21). This principle was first explained in Werden (1996). It was further elaborated in Farrell and Shapiro (2010), who introduced the concept of “Upward Pricing Pressure” (UPP).

³ Two seminal articles are Farrell and Shapiro (1990) and Werden (1996). The first analyzed horizontal mergers with Cournot competition and identical products, while the second analyzed horizontal mergers in a setting with Bertrand competition and differentiated products.

⁴ See, for example, Crooke et al. (1999), who demonstrate the very diverse predictions for post-merger price increases depending on the choice of a linear, logit, almost-ideal demand system (AIDS), or isoelastic demand function.

prices charged by upstream to downstream firms and the downstream firms' prices to the final consumers. The reason is that changes in the bargaining power due to an upstream merger may lead to changes in the fixed transfers between the upstream and the downstream firms rather than in the marginal wholesale prices, thereby providing less incentive for the downstream firms to change the prices offered to the final consumers.

In this article we apply a model with a successive oligopoly. The downstream firms set linear prices to the end-users, while the upstream and downstream firms bargain over non-linear contracts. Contracts between upstream and downstream firms are unobservable. The non-linear contract could take many forms, such as a rebate scheme. To simplify, without loss of generality, we consider a two-part tariff consisting of a price per unit and a fixed fee. In this setting, we show theoretically that an upstream merger will not have any effect on the prices per unit paid by downstream firms (*i.e.*, the marginal wholesale prices) but only on the fixed fees paid to the merging firms. The intuition is that unobservable contracts make it unprofitable to deviate from a marginal wholesale price equal to marginal cost, and that therefore all bargaining power stemming from the upstream merger will be transmitted into changes in the merging parties' fixed wholesale fees. When the downstream firms' input price on the last unit is not changed following an upstream merger, the downstream firms have no incentive to change the prices they charge the consumers.

The theoretical model is simple, but it captures the idea that an upstream merger may lead mainly to changes in non-marginal wholesale fees. We test these predictions on data for an upstream merger in the egg industry in Norway in 2005. The egg producer Prior acquired Norgården, and the merged entity increased its market share for eggs in Norway from 61% to 74%, for deliveries to the four grocery chains selling eggs to final consumers.⁵ Norway imposes heavy constraints on egg imports, so according to the traditional approach to horizontal mergers, such a merger would have anti-competitive effects. The Norwegian Competition Authority blocked the merger, arguing among other things that it would lead to higher prices to consumers. This decision was overturned after an appeal to the Ministry of Government Administration and Reform. The Ministry accepted that the merger was anti-competitive, but allowed it in order to increase the upstream firms' and, in turn, the farmers' revenues. This therefore presents a unique chance to investigate an upstream merger

⁵ In this article, the terms "acquisition" and "merger" will be used interchangeably.

that the competition authorities claimed would have anti-competitive effects.

We have monthly prices (price indices) for eggs at the wholesale and consumer level for several years before and after the merger in 2005. There was no change in the development of the consumer prices around the time of the merger. This is consistent with our theoretical predictions, but might be explained by no additional exploitation of market power by the upstream firms after the merger. The average wholesale price paid to the merged firm did however increase strongly following the merger. This indicates that the merger did lead to an increase in upstream market power. Since the input in production of chicken meat is very similar to the input in the production of eggs, we also use chicken meat as a control group and perform a difference-in-difference analysis. The analysis confirms our initial findings. First, we find no effect of the upstream merger on consumer prices on eggs compared to the consumer prices on chicken meat. Second, we find a significant effect on the average wholesale prices on eggs compared to the corresponding wholesale prices for chicken meat. The results are consistent with the theoretical prediction that an upstream merger would lead to changes in the fixed transfers, rather than to changes in the marginal wholesale prices. The results are also in line with anecdotal evidence indicating that fixed fees and rebates are used in secret vertical contracts in this particular market. We investigate other possible explanations – such as changes in upstream firms’ costs or changes in retail concentration – but these other factors cannot explain the observed price patterns.

Although upstream mergers are common, the theoretical literature on them is sparse. Some early contributions focused on linear contracts between upstream and downstream firms, and therefore do not capture the main mechanism we focus on in this article.⁶ Inderst and Wey (2003) model non-linear contracts, but they assume that downstream firms are independent and therefore do not capture any possible anti-competitive effects in the downstream market. Milliou and Petrakis (2007) apply a successive duopoly model to study upstream mergers, but they focus on how upstream mergers may influence the choice between two-part tariffs and a linear wholesale price. They assume Cournot competition downstream and contract observability. This leads to marginal wholesale prices below marginal costs, and even more so after an upstream merger. In their main model, unlike in our study, they therefore find that an upstream merger leads to lower consumer prices. Finally, O’Brien and Shaffer (2005) apply a model similar to ours, except for assuming

⁶ See Henrick Horn (1988), Ungern-Sternberg (1996) and Dobson and Waterson (1997).

monopoly in the downstream market. The theoretical model presented here is an extension of their model, and illustrates that their main results extend to a more general setting with several downstream firms and unobservable contracts.⁷

The empirical literature on upstream mergers and how vertical restraints will influence the prices to the final consumers is even more sparse. To our knowledge, Villas-Boas (2007a) and Manuszak (2010) are the only empirical investigations that explicitly analyze the role of vertical restraints for the price effects in the consumer market from an upstream merger.⁸ Neither of these two studies have access to wholesale prices. They estimate consumer demand, make assumptions concerning the tariff between the upstream and the downstream firms, and then use their demand estimates to simulate the effects of an upstream merger on consumer prices.⁹ In contrast to these articles, we have data for wholesale prices and can test how the upstream merger affected average wholesale prices.

The article is organized as follows. In the next section the institutional setting for the upstream merger in the Norwegian market for eggs is presented. In Section 1.3 a theoretical model is applied to formulate testable predictions. Section 1.4 contains the empirical study of the merger in question. Section 1.5 discusses the robustness of the findings, while Section 1.6 concludes.

1.2 Institutional setting for the upstream merger

Eggs in Norway are produced by many farmers. Each farmer delivers eggs to firms that pack and process the eggs, and then sell them to retail chains. We denote these firms “upstream firms,” and we denote the retail chains “downstream firms.” At the retail level, four chains controlled almost 100% of the domestic food market in 2005. There are high duties on imports of eggs to Norway, to protect Norwegian

⁷ O’Brien and Shaffer (2005) indicate that their results extend to an oligopoly setting downstream, but they do not explicitly model downstream competition.

⁸ There are other empirical studies of mergers that do not discuss vertical structure, but which nevertheless can shed light on whether there is any price effect of an upstream merger that is passed on to final consumers. See, for example, Ashenfelter and Hosken (2010). They analyze five upstream mergers in the US and find that four of the five resulted in higher prices to final consumers. See also Ashenfelter, Hosken, and Weinberg (2013), who study the price effect for final consumers of the Maytag-Whirlpool merger.

⁹ They focus mainly on situations with a linear contract between upstream and downstream firms. There are studies that use estimated consumer demand to investigate the form of the contract between upstream and downstream firms, for example whether it is non-linear (see, for example, Villas-Boas (2007b) and Bonnet and Dubois (2010)). However, these articles do not analyze the effects of upstream mergers.

farmers and allow them to receive higher revenues.¹⁰

The largest upstream firm, Prior, was a farmer-owned cooperative, where the farmers owned the firm that packed, processed and resold the eggs.¹¹ Prior had been appointed by the government to take a role as a market regulator. Its role was to influence total supply in the industry in such a way that a target average wholesale price was reached (the “målpris”). This target price was at the time of the acquisition set in the annual negotiations between the farmers’ associations and the government.¹² Prior had the authority to enforce measures that would dampen the farmers’ incentives to increase egg production. It could also start exports, to overcome a short run excess supply of eggs in the domestic market. However, Prior had no direct control over other upstream firms’ total production, nor over the prices they offered to downstream firms. It is reason to believe that some of the measures – for example Prior’s obligation to receive deliveries from other domestic egg producers and to start exports when there was excess supply – might have given incentives for other firms in this industry to expand their production and in that respect undermine Prior’s attempts to achieve the target price.

Prior, with more than 60% of the total supply in the Norwegian market, announced on March 31, 2005 that they would acquire Norgården, the second largest supplier with approximately 15% of total supply. In their “efficiency defense” for the acquisition, they claimed that the acquisition would enable them to regulate the market more efficiently:¹³

The parties claim that it is crucial that Prior as a market regulator control the supply from a sufficiently large number of upstream firms to ensure an efficient control over the total supply in Norway (our translation).¹⁴

¹⁰ The information in this section builds on the public decision letter of September 29, 2005 by the Norwegian Competition Authority (Konkurransetilsynet) concerning Prior’s acquisition of Norgården.

¹¹ Prior later merged with the meat producer Gilde, which was also a cooperative. The merged entity was named Nortura.

¹² The target price was an important element in the annual agreement between the farmers’ associations and the government (Jordbruksavtalen). They agreed on how much direct support the farmers should receive, and how much the farmers should receive through high prices in the market. The latter was achieved by trying to reach the target price.

¹³ Note that Norway uses a “total welfare” standard in merger control, which means that all kinds of efficiency gains should be taken into account. In addition to the argument put forward here, they also claimed that the acquisition would lead to savings in fixed costs.

¹⁴ See the Norwegian Competition Authority’s decision letter from September 29, 2005 on Prior’s acquisition of Norgården, p. 32.

This indicates that a motivation for the acquisition was to better control total supply, in other words to strengthen the market power of the merging parties. This would in turn enable the merged firm to extract a higher average price from the downstream firms. In Figure 1.1 Prior's market share before and after the merger is shown.

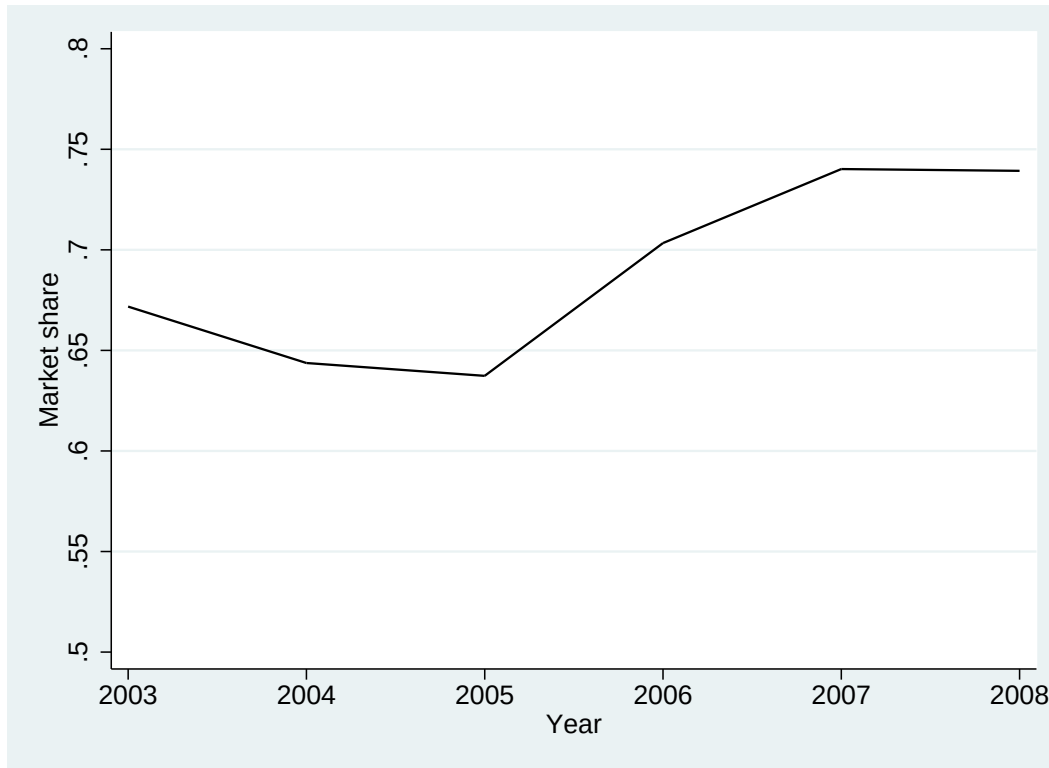


Figure 1.1: Prior's market share.
Source: Statens Landbruksforvaltning.

As seen from the figure, Prior's market share was decreasing before the merger. After the merger the market share increased by approximately 10 percentage points. The fall in the market share before the merger indicates a potential competitive problem, where Prior as the market regulator was not able to control the quantity supplied by other upstream firms. In Figure 1.2 both the target price ("måpris") and the actual price are shown, as a solid and a dotted line respectively. The first is the target for the average wholesale price received by Prior, while the second is the actual average wholesale price.



Figure 1.2: Target price and realized price for eggs, delivered by Prior.
Source: Statens Landbruksforvaltning.

Figures 1.1 and 1.2 seen in combination suggest that Prior as a market regulator did have problems before the merger in controlling the total supply. In a situation with excess supply in the industry, Prior as a market regulator had an obligation to receive deliveries from other producers of eggs. For example, in 2004 it received deliveries from other producers that amounted to approximately 4% of total production.¹⁵ In 2004 and 2005 the market regulator did implement measures, including exporting at a loss, that probably helped to reduce the divergence between the target price and the actual price in the domestic market during that particular period.

In September 2005, the Norwegian Competition Authority blocked the merger, claiming that it would be anti-competitive:

The Norwegian Competition Authority has found that the acquisition

¹⁵ See Statens Landbruksforvaltning (2005, p. 16). Prior received 2,100 tonnes of eggs from other producers that year. Total domestic production was slightly less than 50,000 tonnes.

will lead to a substantial lessening of competition in the relevant markets. [...] We therefore find that the acquisition will lead to higher prices, to the detriment of consumers (our translation).¹⁶

However, in their decision letter there is no discussion of the vertical structure in this market. From this we can infer that they assumed this upstream merger would lead to higher wholesale prices that would – at least to some extent – be passed on to the final consumers.

The merging parties appealed the decision to the Ministry of Government Administration and Reform. In their February 6, 2006 decision, the Ministry supported the analysis of the Norwegian Competition Authority, and found that the acquisition would lead to a substantial lessening of competition. Despite this, the Ministry overturned the decision and approved the acquisition even though they expected it to be anti-competitive. They took into consideration the goals of the government’s agricultural policy:

The Ministry finds that the acquisition of Norgården is of importance for the goal of the agricultural policy to protect the Norwegian production of eggs and revenues for the producers of eggs (our translation).¹⁷

A survey done by the Norwegian Competition Authority in 2005 found that fixed fees and various kinds of rebates (*i.e.*, non-linear tariffs) are quite common in contracts between retail chains and their suppliers.¹⁸ This strongly suggests that fixed fees are also present in this particular industry. The Norwegian Competition Authority also found that both upstream and downstream firms treat each contract as confidential information, so that rivals cannot observe, for example, rebates and fixed fees in other firms’ contracts.¹⁹ In line with this, it is no surprise that it is hard

¹⁶ See the Norwegian Competition Authority’s decision letter from September 29, 2005 on Prior’s acquisition of Norgården, page 31.

¹⁷ See the decision made by the Ministry on February 6, 2006, p. 23. According to § 21 in the Norwegian competition law, the Ministry has the option to allow a merger even if it is anti-competitive if it can claim that there are other reasons for accepting it.

¹⁸ In Norwegian Competition Authority (2005, p. 45) they state the following concerning fixed fees in general: “Our survey shows that fixed fees are used to a large extent in the markets we have investigated” (our translation).

¹⁹ See Norwegian Competition Authority (2005, p. 55). They found that in some particular industries the rival firms have knowledge about what they call “the main elements of the contract.” However, even in this case the contract could be regarded as unobservable, since for example the rival firm does not exactly observe the marginal wholesale price.

to get exact knowledge about the specific contracts between upstream and downstream firms in this industry. All in all, this suggests that in this particular industry the firms use non-linear contracts, including fixed fees, that are unobservable. This motivates our chosen modeling approach, with differentiated firms at upstream and downstream level and secret non-linear contracts.²⁰

1.3 Theoretical predictions

O'Brien and Shaffer (2005) analyze a setting with an oligopoly upstream and a monopoly downstream, with non-linear tariffs between the differentiated upstream firms and the downstream monopolist. According to the description in the previous section, there are unobservable non-linear contracts between (differentiated) upstream and downstream firms in the particular successive oligopoly we investigate. These features are identical with the assumptions made in O'Brien and Shaffer (2005), except for their assumption of a downstream monopoly. It is therefore natural to extend the model in O'Brien and Shaffer (2005) to a successive oligopoly with unobservable contracts, and apply this model to predict the price effects of an upstream merger in our particular market.²¹

1.3.1 The model and pre-merger equilibria

Let $N \geq 2$ single-product producers each supply a differentiated product to a set of $M \geq 1$ differentiated retailers. If every retailer carries every product, the consumers must choose how much to buy of product i ($i = 1, \dots, N$) at retailer r ($r = 1, \dots, M$), meaning that they choose between $N \cdot M$ product/retailer combinations. The demand for product i at retailer r is given by

$$q_{ir} = D_{ir}(\mathbf{p}), \quad (1.1)$$

where $\mathbf{p}$ is a vector of all consumer prices. Assume that demand for product i at retailer r is decreasing in the price of i at r , and increasing in all other consumer prices. Own-price effects are assumed to dominate cross-price effects, meaning that demand is reduced when all prices increase by the same amount.

²⁰ One may think that eggs produced by different producers constitute a homogeneous product. However, in the Norwegian market there are relatively large price differences for different brands of eggs, indicating that the consumers view the brands of different producers as differentiated.

²¹ The model and results presented are borrowed from Ulsaker (2013).

The structure of the game is as follows: In the first stage, each producer negotiates with each retailer over a supply contract. The tariff between producer i and retailer r takes the following form: $T_{ir}(q_{ir}) = w_{ir}q_{ir} + f_{ir}$. All $N \cdot M$ negotiations are assumed to take place simultaneously and independently, and each party in a given negotiation takes the outcome of the other $N \cdot M - 1$ negotiations as given.²² In the second stage, the multi-product retailers compete in prices, demands are satisfied and payments are made in accordance with contracts. Let the contracts of rival retailers be unobserved throughout the game. This means that the prices set by retailer r is independent of the outcome of the negotiations between the producers and the rival retailers. On the other hand, the quantity q_{ir} actually sold will depend on all wholesale prices. Assume that for every vector of wholesale prices $\mathbf{w}$, there is a unique vector of consumer prices, $\mathbf{p} = \mathbf{P}(\mathbf{w})$.²³ This makes it possible to write the demand at retail level as $q_{ir} = Q_{ir}(\mathbf{w})$. As with consumer prices, assume that demand for product i at retailer r is decreasing in own wholesale price and increasing in the other wholesale prices, and let own-price effects dominate cross-price effects.

The outcome of the negotiations between producer i and retailer r is assumed to be the solution to $\max_{w_{ir}, f_{ir}} N_{ir}$, where

$$N_{ir} = \gamma^{ir} \ln(\pi_i - \nu_i^r) + (1 - \gamma^{ir}) \ln(\pi_r - \nu_r^i). \quad (1.2)$$

In this equation, $\gamma^{ir} \in (0, 1)$ represents the bargaining power of the producer in this particular negotiation. π_i and π_r are the profits of the producer and retailer, respectively, and ν_i^r and ν_r^i their disagreement payoffs. In this negotiation, it is not the profit that stems from the sale of product i at retailer r that a firm seeks to maximize, but the firm's total profit. The profits in the objective function are therefore:

$$\pi_i = Q_{ir}(\mathbf{w})(w_{ir} - c) + f_{ir} + \sum_{s=1, s \neq r}^M (Q_{is}(\mathbf{w})(w_{is} - c) + f_{is}) \quad (1.3)$$

²² The assumption of simultaneous and independent negotiations is also made in Inderst and Wey (2003) and in Milliou and Petrakis (2007). One interpretation is that each firm sends one representative to each of the negotiations the firm participates in, and that representatives are not able to coordinate their actions.

²³ Since the prices at retailer r only are affected by its own wholesale prices, we could write $p_{ir} = P_{ir}(\mathbf{w}_r)$, where $\mathbf{w}_r$ is the wholesale prices of retailer r .

and

$$\pi_r = Q_{ir}(\mathbf{w})(P_{ir}(\mathbf{w}) - w_{ir}) - f_{ir} + \sum_{j=1, j \neq i}^N (Q_{jr}(\mathbf{w})(P_{jr}(\mathbf{w}) - w_{jr}) - f_{jr}). \quad (1.4)$$

Assume that if the parties fail to agree on a contract, f_{ir} is set to zero and w_{ir} to infinity, meaning that retailer i will choose not to stock product i . The disagreement profits can then be written as:

$$\nu_i^r = \sum_{s=1, s \neq r}^M (Q_{is}(w_{ir} = \infty, \mathbf{w}_{-ir})(w_{is} - c) + f_{is}) \quad (1.5)$$

and

$$\nu_r^i = \sum_{j=1, j \neq i}^N (Q_{jr}(w_{ir} = \infty, \mathbf{w}_{-ir})(P_{jr}(w_{ir} = \infty, \mathbf{w}_{-ir}) - w_{jr}) - f_{jr}), \quad (1.6)$$

where $\mathbf{w}_{-ir}$ is shorthand for the remaining wholesale prices.

A bargaining equilibrium is a set of contracts that solves $\max_{w_{ir}, f_{ir}} N_{ir}$ for every pair of producer and retailer, and where in each negotiation the equilibrium outcomes of the other negotiations are treated as given. This equilibrium concept focuses on pairwise deviations. Since the parties in a given negotiation take the outcomes of the $N \cdot M - 1$ other negotiations as given, they will have a common interest in maximizing their joint profit and distributing it through the fixed fee. The following proposition applies:

Proposition 1. *When $N \geq 2$ producers and $M \geq 1$ retailers negotiate simultaneously and pairwise over a two-part tariff, each negotiation yields an effective solution, in the sense that the marginal wholesale prices facilitate maximization of the joint profit of the negotiating parties, given the expected outcome of the other negotiations. This again implies that the marginal wholesale prices are equal to marginal cost in equilibrium.*

Proof. See Appendix A.

Since marginal wholesale prices equal marginal cost in equilibrium, the profit of the producers must stem only from the fixed fees. In equilibrium, the fixed fee paid by retailer r to producer i is:

$$f_{ir}^* = \gamma^{ir} \left(\begin{aligned} & Q_{ir}(\mathbf{w}^*) (P_{ir}(\mathbf{w}^*) - c) + \sum_{j=1, j \neq i}^N Q_{jr}(\mathbf{w}^*) (P_{jr}(\mathbf{w}^*) - c) \\ & - \sum_{j=1, j \neq i}^N Q_{jr}(w_{ir} = \infty, \mathbf{w}_{-ir}^*) (P_{jr}(w_{ir} = \infty, \mathbf{w}_{-ir}^*) - c) \end{aligned} \right) \quad (1.7)$$

As the bargaining power of the producer (γ^{ir}) approaches zero, this fixed fee approaches zero. As γ^{ir} approaches one, the fixed fee approaches the difference between retailer r 's profit (gross of fixed fees) when selling product i and when not selling product i . This difference will be strictly positive given that the products are not perfect substitutes.

1.3.2 Post-merger equilibrium

Assume now that producers 1 and 2 merge. Let the merged firm be denoted u . The two-product merged firm negotiates with retailer r over the terms in a contract of the form $T_{ur}(q_{1r}, q_{2r}) = w_{1r}q_{1r} + w_{2r}q_{2r} + f_{ur}$. The non-merging producers negotiate with the retailers as they did before the merger, and the timing of the game is as in the previous subsection. Let the merged firm and retailer r solve $\max_{w_{1r}, w_{2r}, f_{ur}} N_{ur}$, where

$$N_{ur} = \gamma^{ur} \ln(\pi_u - \nu_u^r) + (1 - \gamma^{ur}) \ln(\pi_r - \nu_r^u). \quad (1.8)$$

The profit functions in the objective function are given by:

$$\pi_u = \sum_{i=1}^2 Q_{ir}(\mathbf{w}) (w_{ir} - c) + f_{ur} + \sum_{s=1, s \neq r}^M \left(\sum_{i=1}^2 Q_{is}(\mathbf{w}) (w_{is} - c) + f_{us} \right) \quad (1.9)$$

and

$$\pi_r = \sum_{i=1}^2 Q_{ir}(\mathbf{w}) (P_{ir}(\mathbf{w}) - w_{ir}) - f_{ur} + \sum_{j=3}^N (Q_{jr}(\mathbf{w}) (P_{jr}(\mathbf{w}) - w_{jr}) - f_{jr}). \quad (1.10)$$

The disagreement profits are now given by:

$$\nu_u^r = \sum_{s=1, s \neq r}^M \left(\sum_{i=1}^2 Q_{is}(w_{1r} = w_{2r} = \infty, \mathbf{w}_{-1r,2r}) (w_{is} - c) + f_{us} \right) \quad (1.11)$$

and

$$\nu_r^u = \sum_{j=3}^N (Q_{jr}(w_{1r} = w_{2r} = \infty, \mathbf{w}_{-1r,2r}) (P_{jr}(w_{1r} = w_{2r} = \infty, \mathbf{w}_{-1r,2r}) - w_{jr}) - f_{jr}). \quad (1.12)$$

The maximization problems in the negotiations are analogous to the ones analyzed in the previous subsection. The parties of a given negotiation still take the outcome of the other negotiations as given, and still have a common interest in maximizing joint profit. The following applies:

Proposition 2. *When two of the producers merge, each negotiation still yields an effective solution. The marginal wholesale prices will not be affected and will still be equal to marginal cost.*

Proof. See Appendix A.

The equilibrium fixed fee paid by retailer r to the merged firm is given by:

$$f_{ur}^* = \gamma^{ur} \left(\begin{aligned} & \sum_{i=1}^2 Q_{ir}(\mathbf{w}^*) (P_{ir}(\mathbf{w}^*) - c) + \sum_{j=3}^N Q_{jr}(\mathbf{w}^*) (P_{jr}(\mathbf{w}^*) - c) \\ & - \sum_{j=3}^N Q_{jr}(w_{1r} = w_{2r} = \infty, \mathbf{w}_{-1r,2r}^*) (P_{jr}(w_{1r} = w_{2r} = \infty, \mathbf{w}_{-1r,2r}^*) - c) \end{aligned} \right). \quad (1.13)$$

The expression in the parentheses on the right-hand side of (1.13) is the incremental profit (gross of fixed fees) that retailer r earns by selling product 1 and 2, compared to not selling any of them, or equivalently, the loss of gross profit that retailer r would suffer if product 1 and 2 were simultaneously made unavailable. Prior to the merger, the merged firms earned $f_{1r}^* + f_{2r}^*$ through sales at retailer r . f_{1r}^* corresponds to the fraction γ^{1r} of the loss in gross profit incurred by retailer r in the case that product 1 became unavailable at the retailer, given that product 2 would still be available at retailer r .

The merger is profitable for the merging firms if $f_{ur}^* > f_{1r}^* + f_{2r}^*$. Assuming that $\gamma^{1r} = \gamma^{2r} = \gamma^{ur}$, this would be the case if the loss (in terms of gross profit) for the retailer is greater when products 1 and 2 are simultaneously made unavailable than the sum of the losses of not being able to stock product 1 (when product 2 is available) and of not being able to stock product 2 (when product 1 is available). Since the products are substitutes, this condition will be met. In other words, the outside options for each retailer are less profitable after the acquisition, since both product 1 and 2 will be unavailable if a retailer fails to sign a contract with the merged firm. The increased upstream market power following the acquisition leads to higher fixed fees paid from the retailers to the merging firms.

In this model, an upstream merger leads to no changes in the marginal wholesale price, but only to a change in the fixed wholesale fees for the merging parties. It implies that the *average* wholesale price (*i.e.*, total payment divided by quantity) will increase for the merging parties. Since there are no changes in the marginal wholesale prices, the model predicts no change in consumer prices. The predictions which can be taken to data is then the following:

Proposition 3. *The upstream merger is expected to lead to (i) higher average wholesale prices paid to the merged firm, and (ii) no changes in consumer prices.*

Several assumptions might be crucial for the results reported in Proposition 3. First, we assume that after the merger, the merging firm bargains with the downstream firms over both products at the same time (bundling). If the firms instead negotiate separately over each of the merged firm's two products (no bundling), then it can be shown that the upstream merger could lead to higher marginal wholesale prices.²⁴ Second, we assume contract unobservability. If contracts were observable, marginal wholesale prices could differ from marginal costs both before and after the merger (to dampen competition at the retail level). Hence, it is an empirical question whether consumer prices increase or not following an upstream merger.

Our theoretical model illustrates that an upstream merger may well lead to an increase in market power and average wholesale prices, without leading to changes in the prices at consumer level. The prediction we want to test in our empirical analysis is that the merger in question did not affect the consumer prices, but that it led to higher average wholesale prices than what would have been the case absent

²⁴ This is shown in O'Brien and Shaffer (2005) in a setting with monopoly downstream, and in Ulsaker (2013) in a similar setting except for oligopoly downstream.

the merger. Since there may have been underlying trends in the prices before the merger occurred, what we are interested in is whether the merger led to deviations from the pre-merger trends at wholesale and at consumer level.

1.4 Empirical analysis

The primary focus of this article is the possible effects of the 2005 merger on egg prices. In contrast to other empirical studies, we have data on both the wholesale and the retail level. Chicken meat will be used as a control group, after a first examination of the changes in the prices of eggs following the acquisition.

1.4.1 Data

The empirical analysis is based on a set of price indices. The price indices for eggs are published by Norwegian Agricultural Economics Research Institute (NILF). “Consumer price” (1998 = 100) is the average price of all eggs sold in grocery stores, counting eggs produced by both the merging parties and other producers. “Average wholesale price” (1998 = 100) is total payment divided by quantity delivered, and uses a weighted average of all types of eggs, except for organic eggs, delivered from two of Prior’s plants. While the baskets of eggs used to construct the two price indices are not identical, the degree of correspondence is nevertheless high, since organic eggs account for a small proportion of total egg sales, and since Prior’s market share was between 60% and 75% in the period under consideration.

We will in the empirical analysis compare egg prices with chicken meat prices. The price indices for chicken meat are published by NILF (1998 = 100). As control variables we include price data for a range of important ingredients in poultry feed. These data are available from the Food and Agriculture Organization of the United Nations. Data on electricity prices are provided by Nord Pool Spot, while the index of consumer food prices we use is published by NILF.

1.4.2 Prices on eggs

Figure 1.3 shows price indices for consumer prices and average wholesale prices for eggs. As a reference, we also plot an index for consumer food prices.

Looking at Figure 1.3, there is no obvious break in the consumer price series for eggs around the time of the merger. Furthermore, the growth in the prices at

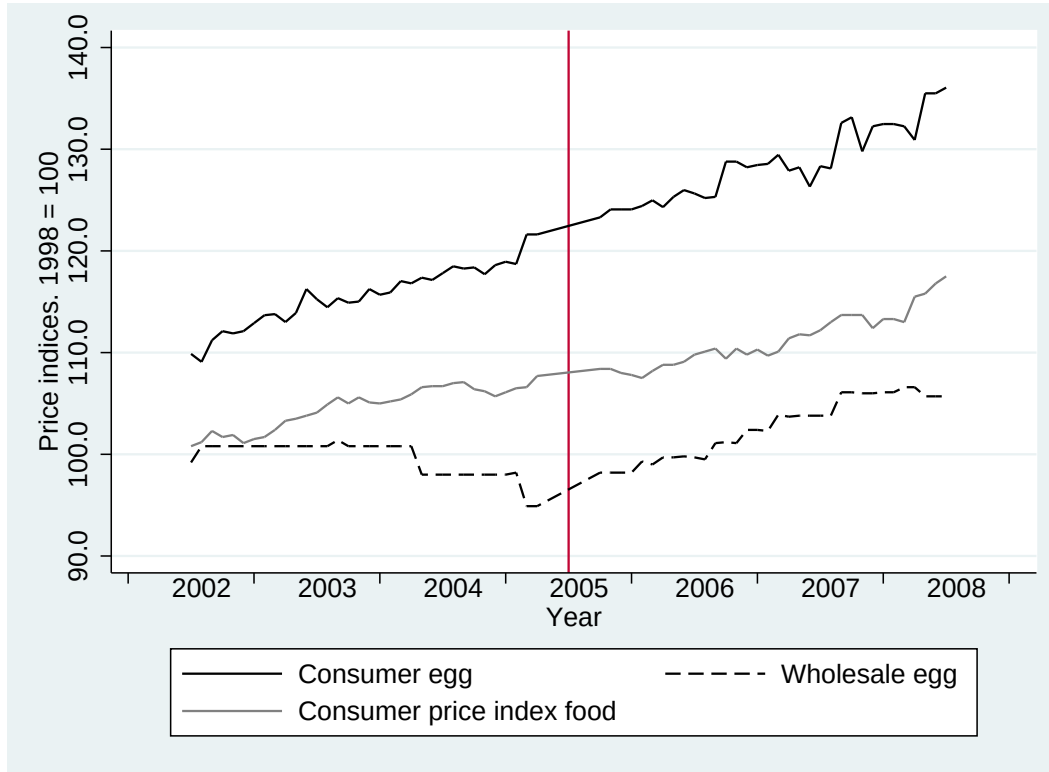


Figure 1.3: Price indices of consumer price and average wholesale price for eggs, and consumer prices for food. 1998 = 100. Vertical line indicates time of merger (2005m5).

Source: NILF.

consumer level seems to follow the same pattern as the general growth in consumer food prices. Considering the wholesale prices, there does seem to be a break in the series around the time of the merger.

Since the primary focus of the article is on the effects of the 2005 merger on wholesale and consumer prices, we focus on the period between May 2002 and May 2008, that is, three years before and three years subsequent to the merger. This makes it possible to compare the development in the price series in the years leading up to the merger with development after the merger, without having to take into account changes and shifts in the price series occurring several years before or after the merger, which are unlikely to be directly related to the merger.

As a first empirical exercise, we examine the wholesale price using a regression model where an exponential time trend is included, that is, allowing for constant

price growth from one month to the next. In addition we look for a structural break at the time of the official merger, May 2005, by allowing for both a change in the time trend and a potential shift in the price series at this date. The equation we estimate is the following:²⁵

$$\begin{aligned} \ln P_t^{wholesale} = & \beta_0^{wholesale} + \beta_1^{wholesale} \cdot D(t \geq 2005m5) \\ & + \gamma_0^{wholesale} \cdot t + \gamma_1^{wholesale} \cdot D(t \geq 2005m5) \cdot t \\ & + \beta_2^{wholesale} \cdot X_t + u_t^{wholesale} \end{aligned} \quad (1.14)$$

Figure 1.3 indicates that there was a change in the development of wholesale prices occurring around the time of the merger. The apparent change could of course be driven by factors unrelated to the merger, for example cost changes. To account for this, log-transformed price indices for soybeans and soybean oil, together with log-transformed price indices for wheat and maize are included as control variables. All these factors are important ingredients in the feed of egg-laying poultry. We also include an index of electricity prices, as well as monthly dummies. The control variables are gathered in the vector X_t .

We exclude the month of the merger, in addition to two observations before the merger and two observations after the merger, to avoid having to make a stand on exactly when the merging parties start to coordinate their behavior.²⁶

To account for potential autocorrelation in the error terms, $u_t^{wholesale}$, the model is estimated using the Prais-Winsten GLS method. The results are reported in Table 1.1, Column 1.²⁷ There is a negative trend before the merger ($\hat{\gamma}_0^{wholesale} = -0.0015$), which corresponds to a yearly growth rate of -1.8% . The coefficient $\hat{\gamma}_1^{wholesale} = 0.0036$ is highly significant and positive, indicating increased growth rate after the merger. The estimated post-merger trend is $\hat{\gamma}_0^{wholesale} + \hat{\gamma}_1^{wholesale} = 0.0021$. This corresponds to a annualized growth rate of 2.5% . The annualized growth rate is thus 4.3% higher after the merger than before the merger. A higher growth rate suggests that prices increased gradually after the merger, rather than making a

²⁵ The time variable used in this and all following regressions is a transformed variable indicating the number of months after the merger. This enables us to interpret $\beta_1^{wholesale}$ in Equation 1.14 as the shift in the price series at the time of the merger.

²⁶ See Ashenfelter and Hosken (2010) for further discussion.

²⁷ When estimating the model using OLS, autocorrelation is strongly indicated by a Durbin-Watson d-statistic of 0.61.

discrete jump. This can be explained by the fact that wholesale contracts may have considerable length, and that not all contracts between the merging firm and its buyers were renegotiated at the same time.

Turning now to the consumer prices, an equation corresponding to Equation 1.14 is estimated. A possible effect of the merger on the consumer prices could of course have been offset by changes in other variables affecting consumer prices. Therefore, a consumer food price index and an electricity price index are included as controls in the regression together with monthly dummies.

The results are summarized in Table 1.1, Column 2. The estimated trend before the merger is 0.0018, which corresponds to a yearly growth rate of 2.2%. We observe a small and statistically insignificant change in the time trend, $\hat{\gamma}_1^{consum} = 0.0001$. The estimated shift in the price series at the time of the merger ($\hat{\beta}_1^{consum} = 0.0098$) is also small and statistically insignificant. Thus, there are no indications of a break in the consumer prices at the merger point.

The results reported in Columns 1 and 2 of Table 1.1 are consistent with the merger having a positive effect on the average wholesale prices, while leaving the consumer prices unchanged. The strong increase in the growth rate at the wholesale level after the merger is not passed through to the consumer prices, which shows no signs of changing at the time of the merger.

VARIABLES	(1) lnP(wholesale egg)	(2) lnP(consumer egg)	(3) lnP(wholesale chicken)	(4) lnP(consumer chicken)
t	-0.0015** (0.0005)	0.0018** (0.0005)	-0.0004** (0.0001)	-0.0019 (0.0010)
D($t \geq 2005m5$)	0.0125 (0.0133)	0.0098 (0.0067)	0.0023 (0.0055)	-0.0197 (0.0124)
D($t \geq 2005m5$)*t	0.0036** (0.0007)	0.0001 (0.0003)	0.0012** (0.0004)	0.0006 (0.0005)
ρ (Autocorrelation coefficient)	0.686	0.274	0.544	0.231
<i>Control variables</i>				
Soybean	YES	NO	YES	NO
Wheat	YES	NO	YES	NO
Maize	YES	NO	YES	NO
Electricity	YES	YES	YES	YES
Monthly dummies	YES	YES	YES	YES
Retail food price index	NO	YES	NO	YES

Robust standard errors in parentheses

** p<0.01, * p<0.05

Observations from May 2002 until May 2008 are used.

Five observations around the time of the merger are excluded.

Table 1.1: Estimation results for eggs and chicken.
Prais-Winsten AR(1) estimation.

1.4.3 Difference-in-difference analysis

Even if we have tried to control for confounding factors by including control variables, it could still be the case that the results discussed above in fact reflect changes in unobservable factors, and not the 2005 merger. To account for this, we compare the development in the egg market with a control group.

The production of chicken meat is a natural control group. The input is very similar, since chicken are used to produce both eggs and chicken meat.²⁸ Moreover, both upstream industries are in Norway heavily protected by import duties and the concentration on the upstream levels are high. In the relevant time period there were no mergers in the chicken meat industry.

The similarity of the two industries, except for the merger in question, makes chicken meat a well suited control group. The (log-transformed) wholesale prices of eggs and chicken meat are shown in Figure 1.4.

As a reference, we estimate regression models for chicken meat prices that are analogous to the ones we estimated for egg prices in the previous subsection. In the difference-in-difference framework, the development of the chicken prices after the merger will be used to estimate the counterfactual development in the egg prices in the absence of the merger, since the chicken prices are assumed to be unaffected by the merger. The results are reported in Column 3 and Column 4 of Table 1.1. We observe that there is a statistically significant increase in the growth rate of chicken prices at wholesale level after the merger, an increase that is not found at consumer level.

Following the literature, we use a standard method to estimate the effects of mergers, and to control for potential confounding factors. This is the difference-in-difference approach, the most common method used to analyze the price effects of mergers.²⁹ We perform a difference-in-difference analysis where we use the price of chicken meat as the control group, and the prices of eggs as the treated group. Equation 1.15 shows the setup used.

²⁸ Chicken farmed for meat are often called broiler chickens, while chicken farmed for eggs are often called egg-laying poultry. Although inputs are almost identical for those two products, there are no economies of scale or scope in production since production of those two products are separated into two different production processes.

²⁹ See for instance Ashenfelter and Hosken (2010), and Ashenfelter, Hosken, and Weinberg (2013).

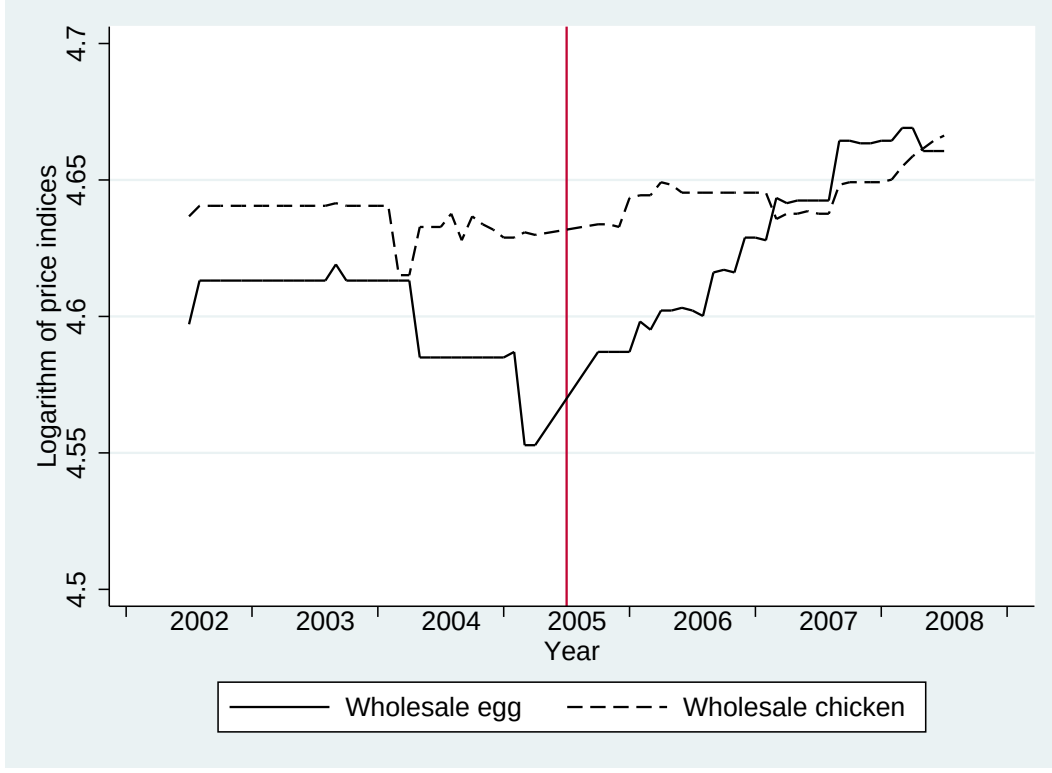


Figure 1.4: Wholesale prices chicken and eggs. Log of index values, 1998 = 100. Vertical line indicates time of merger (2005m5). Source: NILF

$$\begin{aligned}
\ln P_{t,g}^{wholesale-did} = & \beta_0^{wholesale-did} + \beta_1^{wholesale-did} \cdot D(t \geq 2005m5) \\
& + \beta_2^{wholesale-did} \cdot D(egg) + \beta_3^{wholesale-did} \cdot D(t \geq 2005m5) \cdot D(egg) \\
& + \gamma_0^{wholesale-did} \cdot t + \gamma_1^{wholesale-did} \cdot D(t \geq 2005m5) \cdot t \\
& + \gamma_2^{wholesale-did} \cdot D(egg) \cdot t + \gamma_3^{wholesale-did} \cdot D(t \geq 2005m5) \cdot D(egg) \cdot t \\
& + \beta_4^{wholesale-did} \cdot X_t \\
& + \beta_5^{wholesale-did} \cdot D(egg) \cdot X_t + u_{t,g}^{wholesale-did}
\end{aligned} \tag{1.15}$$

As shown in Equation 1.15, we allow for different trends before and after the merger, and different trends for eggs and for chicken meat. We furthermore allow for vertical shifts at the time of the merger. A common assumption in difference-in-difference analysis is that control and treatment groups share a common trend prior

to the treatment. As noted by Angrist and Pischke (2015, p. 197), in a setting such as ours, the treatment effect “comes from sharp deviations from otherwise smooth trends, even where the trends are not common.” Since we have data on prices in several periods before the merger, we can dispense with the assumption of common trends by explicitly modeling separate pre-merger trends for chicken and eggs. Our analysis is therefore valid even when the underlying trends are different for eggs and chicken. Finally we include a vector X_t of control variables, which are the same we used in the previous wholesale level regressions, that is, the log-transformed price indices for soybeans and soybean oil, together with log-transformed price indices for wheat and maize, an index of electricity prices, and finally monthly dummies to control for seasonal effects. We allow the control variables to have separate effects on chicken and egg prices.

The results of this exercise, excluding the month of the merger and the two months immediately before and after, are reported in Table 1.2, Column 1. Note that the coefficients in this column and the coefficients in Column 1 and Column 3 of Table 1.1 can be derived from each other. For example, in Table 1.2, the growth rate in the egg prices at wholesale level before the merger is estimated by $\hat{\gamma}_0^{wholesale-did} + \hat{\gamma}_2^{wholesale-did} = -0.0004 - 0.0011 = -0.0015$, which is identical to the corresponding coefficient in Column 1 of Table 1.1.

We are mainly interested in the coefficients for the interaction terms $D(t \geq 2005m5) \cdot D(egg) \cdot t$ and $D(t \geq 2005m5) \cdot D(egg)$. The first of these measures the difference in the change in the growth rate after the merger, between treatment and control group. The second measures differences in shifts at the merger date, between treatment and control group. The difference-in-difference approach allows us to compute standard errors for these coefficients.

Since the estimated coefficient for $D(t \geq 2005m5) \cdot D(egg)$ is not significantly different from zero, there is no indication of a shift in the egg prices at the time of the merger. However, the coefficient for $D(t \geq 2005m5) \cdot D(egg) \cdot t$, which equals 0.0024, suggests that the merger resulted in a 2.9% increase in the yearly growth rate of the wholesale prices of eggs, compared to the prices of chicken meat. This significant (both economically and statistically) increase in the growth rate again supports the predictions from the theoretical model: The merger seems to have had a positive effect on the average egg prices at the wholesale level.

We are also interested in whether there is a corresponding break in the egg prices at consumer level after the merger. The (log-transformed) consumer prices of eggs

VARIABLES	(1) lnP(wholesale)	(2) lnP(consumer)
t	-0.0004** (0.0002)	-0.0019* (0.0009)
D(t $\geq$ 2005m5)	0.0023 (0.0053)	-0.0197 (0.0145)
D(t $\geq$ 2005m5)*t	0.0012** (0.0003)	0.0006 (0.0005)
D(egg)	-0.2174 (0.2176)	4.4794* (2.2807)
D(egg)*t	-0.0011** (0.0004)	0.0037** (0.0010)
D(t $\geq$ 2005m5)*D(egg)	0.0101 (0.0122)	0.0295 (0.0162)
D(t $\geq$ 2005m5)*D(egg)*t	0.0024** (0.0008)	-0.0005 (0.0006)
ρ (Egg, Chicken)	(0.686, 0.544)	(0.274, 0.231)
<i>Control variables</i>		
Soybean	YES	NO
Wheat	YES	NO
Maize	YES	NO
Electricity	YES	YES
Monthly dummies	YES	YES
Retail food price index	NO	YES

Robust standard errors in parentheses.

** p<0.01, * p<0.05

Observations from May 2002 until May 2008 are used

Five observations around the time of the merger are excluded.

The AR(1) process is assumed to be specific for chicken and eggs.

We allow the control variables to have separate effects on chicken and egg prices.

Table 1.2: Estimation results, Difference-in-Difference.
Prais-Winsten AR(1) estimation.

and chicken meat are shown in Figure 1.5.

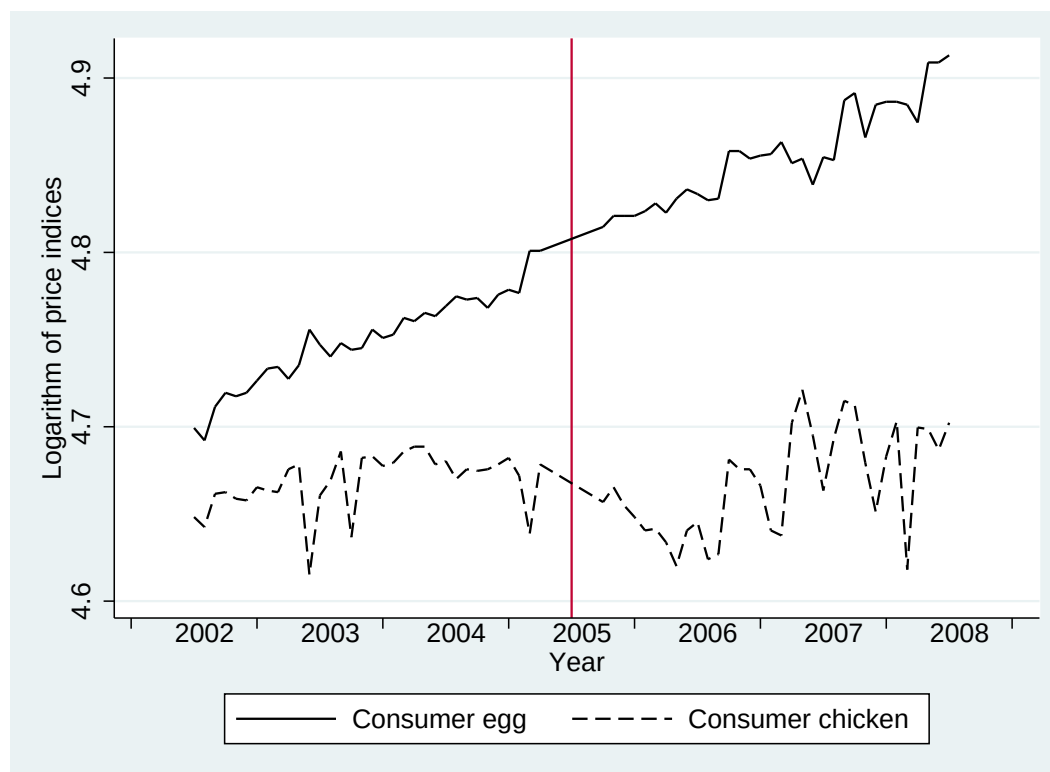


Figure 1.5: Consumer prices chicken and eggs. Log of index values, 1998 = 100. Vertical line indicates time of merger (2005m5).

Source: NILF

The results of a difference-in-difference estimation at the consumer level are reported in Table 1.2, Column 2. Since the estimated coefficient for $D(t \geq 2005m5) \cdot D(egg) \cdot t$ is negative and statistically insignificant, the strong increase in the yearly growth rate of egg prices at the wholesale level (estimated to be 2.9%) does not seem to have been transmitted to the consumer level. Further, the coefficient for $D(t \geq 2005m5) \cdot D(egg)$ is statistically insignificant, meaning that we do not find evidence for a shift in the egg prices at the consumer level. Taken together, the difference-in-difference estimation confirms the conclusions from the previous subsection. The merger led to a significant increase in the average wholesale price, which was not transmitted to the consumer prices. Our interpretation is that while the average wholesale prices increased, the marginal wholesale prices remained unchanged, and that the increased market power of the merging firms was instead exploited through

for example higher fixed fees.

1.5 Discussion

It might be argued that the lack of a structural break in the consumer prices is driven by changes in the market structure at the retail level. In particular, an increase in the competitive pressure at the retail level from 2005 on could explain why the increase in the wholesale prices is not passed on to the consumer prices. Monthly data capturing such changes are hard to get. However, Figure 1.6 reports annual retail concentration, suggesting that it does not seem to be an issue. On the contrary, we observe an increase in concentration on the retail level after 2005.

An alternative way to test the prediction from theory is to look at quantity sold.

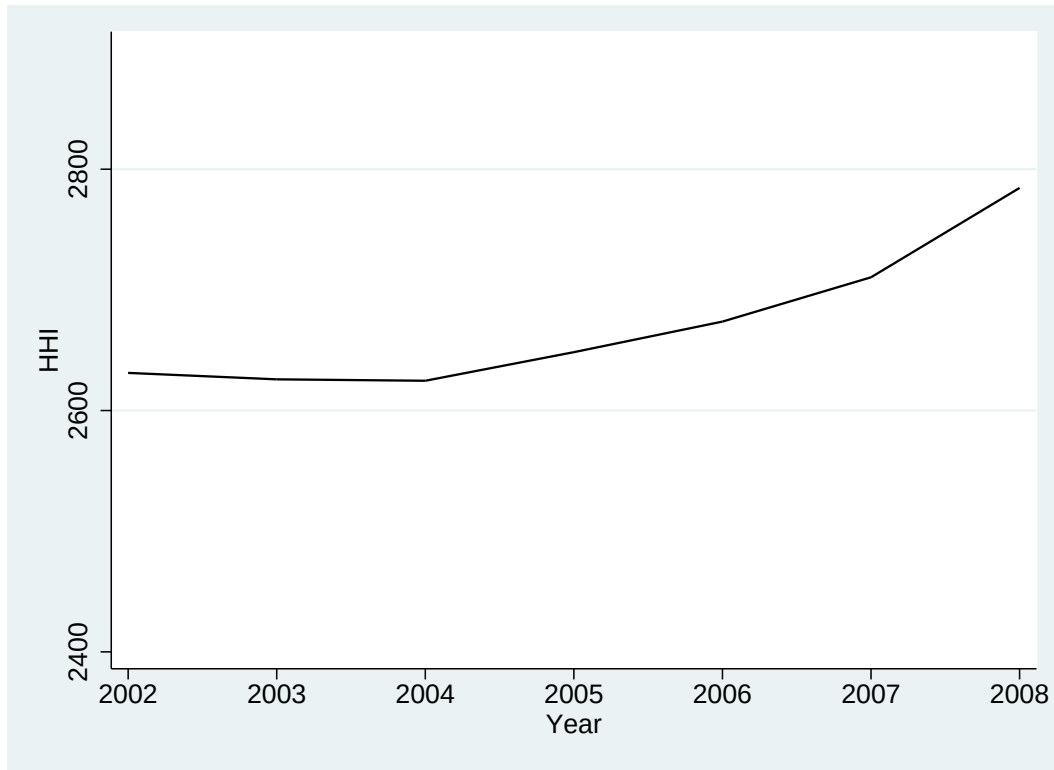


Figure 1.6: HHI retail sector.
Source: AC Nielsen.

Since the model predicts no changes in the marginal price on inputs for the retailers, we expect that the retailers do not change their prices to final consumers following

the merger. One implication of this is that we should observe no change in quantity sold at the time of the merger. Figure 1.7 shows annual sales for eggs and chicken.

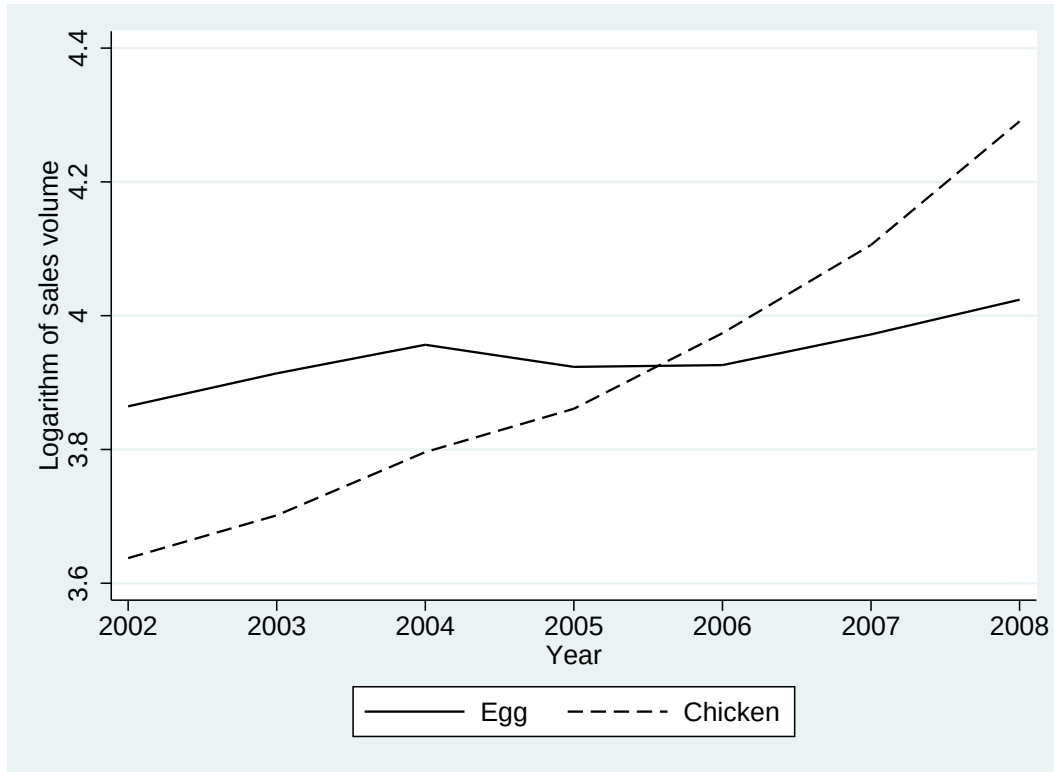


Figure 1.7: Yearly sales volumes of eggs and chicken. Values in 1000 tonnes.
Source: NILF

There seem to be no significant changes in annual egg sales at the time of the merger. If anything, sales volume increased after the merger. This is the opposite of what would be expected if the merger had led to higher prices to final consumers. This piece of evidence is in line with there being no structural break in the final consumer prices. It is also consistent with our interpretation that the shift in the average wholesale prices is caused by shifts in non-marginal payments from downstream to upstream firms.

Looking at the sales volume of chicken in Figure 1.7, we observe an increase over time, also compared to the egg volume. Given the development of the chicken prices and the increasing sales volume, there seems to have been an increase in the demand for chicken in the period considered. Note however that the sales volume is increasing steadily in the sample period, both before and after the merger. Since we

are allowing for different underlying trends in the prices (and thereby demands) of eggs and chicken in the difference-in-difference estimation, the increase in the sales of chicken should not pose a serious challenge to our interpretation of the difference-in-difference estimation. Furthermore, the relative increase in chicken sales can at least partly be explained by the fact that chicken prices decreased relative to egg prices (and relative to the index of food prices) in the period under consideration (see Figure 1.5).

The increased growth in the average wholesale prices might also be explained by measures implemented by the market regulator Prior. Each year, costly efforts ensure high prices, like exporting eggs at a price that leads to a loss for the producers. In Figure 1.8, we show the annual cost of such measures.

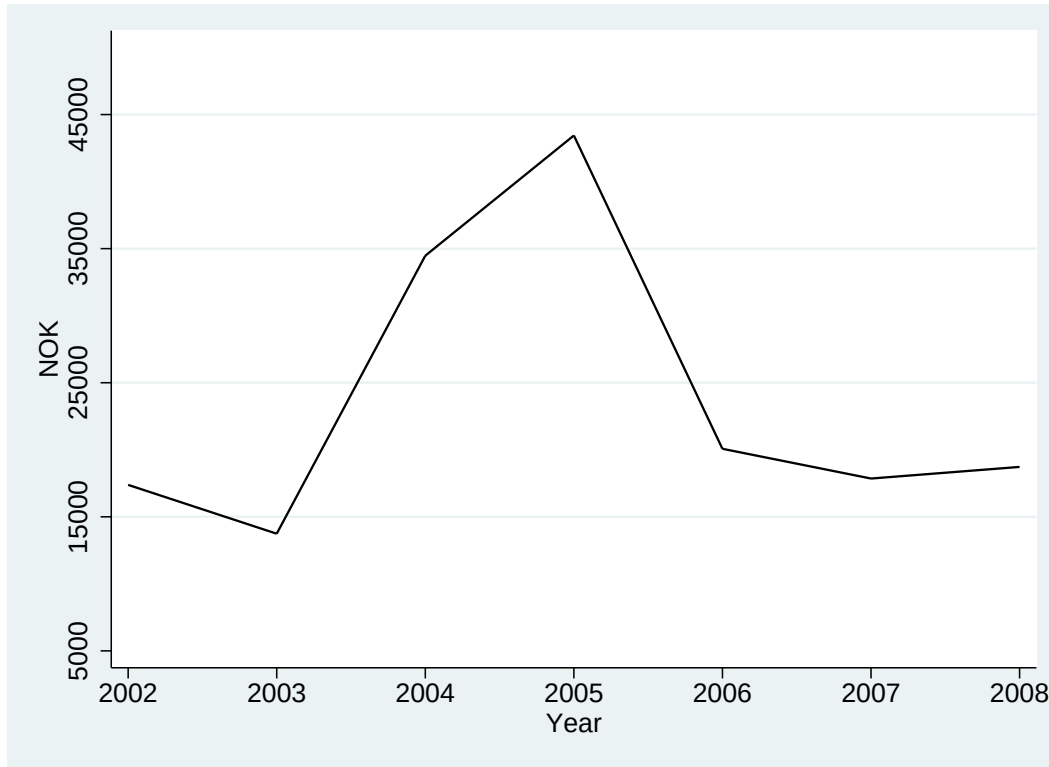


Figure 1.8: Cost of market regulation. Values in thousands.

Source: Annual reports, Statens Landbruksforvaltning (Omsetningsrådet).

This figure shows that the costs associated with market regulation increased in 2004 and 2005, and then dropped again after that. Unfortunately, we do not have monthly data and cannot use this as a control variable in our regressions.

Note, though, that the cost of market regulation drops after the 2005 merger. This indicates that market regulation as such cannot explain the long run trend in prices, where there was a positive shift in the average wholesale price trend after 2005. On the contrary, this indicates that the acquisition made it possible to increase the average wholesale price without extra efforts to regulate the market.

Annual producer prices (transfers paid by the upstream firms to the farmers) are shown in Figure 1.9. Since the upstream acquiring firm is a cooperative owned by its suppliers, the producer price it pays can be seen as a measure of the profits generated by the upstream firm (rather than as a cost to the upstream firm).³⁰ Since the market regulator incurs costs with market regulation, the producer price would then capture the combined effect of the acquisition on average wholesale prices and the costs associated with market regulation. We also have access to monthly data on producer prices.³¹ Using monthly data, we find that the increase in average wholesale prices after the merger is mirrored by an increase in the average producer prices. This indicates that the acquisition led to a combination of higher average wholesale prices and fewer resources spent on market regulation.

We have considered average wholesale prices of the acquiring firm before and after the acquisition. Ideally, we should then compare that with the changes in consumer prices for the acquiring firm's products. Unfortunately, our data are not sufficiently detailed for us to do this. However, the large size of the acquiring firm implies that if there were any change in the acquiring firm's consumer price, it would also be expected to show up in the general consumer price index for eggs we are using. It would also be of interest to check the average wholesale price for the non-merging firms. According to our theoretical model, the acquisition should not affect the average prices of non-merging firms. Unfortunately, we do not have access to detailed wholesale price data for these other firms.

The empirical models in this article are all estimated using price indices. Some might argue that a proper analysis should be based on micro data, such as scanner data, for the products. With scanner data only available at the consumer level one has to make (non-testable) assumptions concerning the price tariff between the

³⁰ In the theoretical model we did not take into account that the upstream firm was a cooperative. However, it does not matter for our theoretical predictions, since a cooperative can be regarded as a firm that is integrated upstream. When negotiating with the retail chains, Prior would be expected to maximize profit in the usual way.

³¹ These monthly producer prices are confidential. Some more details may be given by the authors on request.

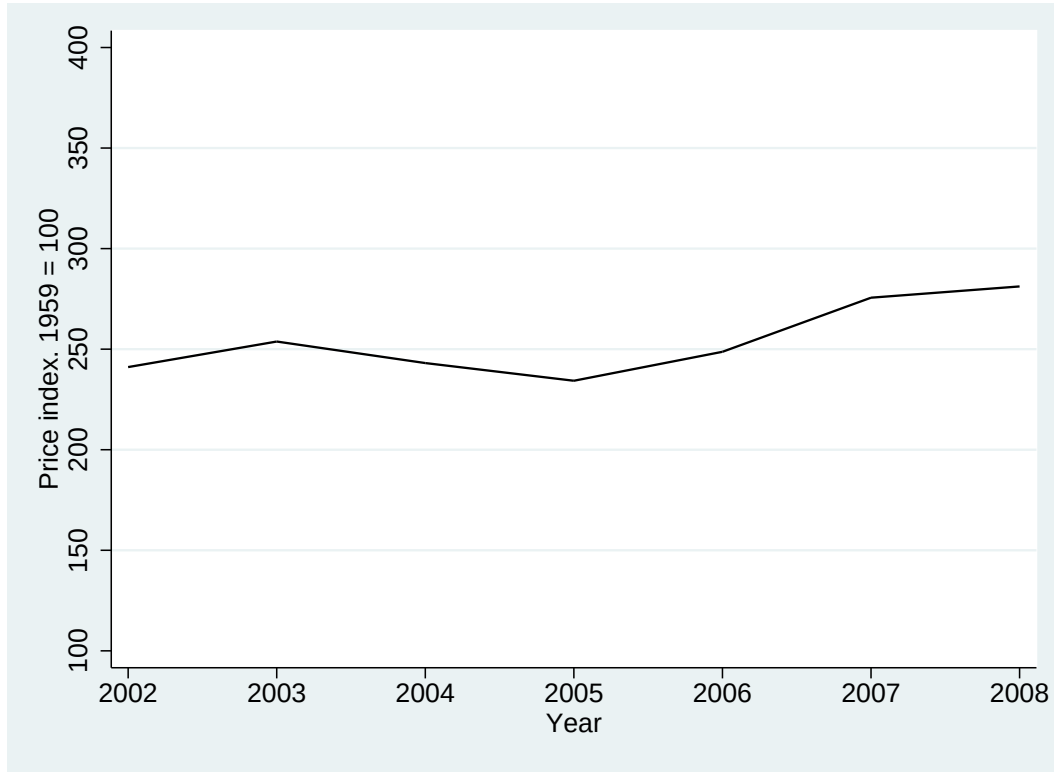


Figure 1.9: Price index of producer price. 1959 = 100.
Source: NILF

upstream and the downstream firms. Furthermore, scanner data are typically used to estimate a system of demand equations such that one can simulate the effects of a merger. Note however that with our data we can observe the price effects of the merger on average wholesale prices directly and do not have to rely on all the necessary assumptions being made when doing merger simulations.

The set of robustness checks we report here, as well as the one we report in Appendix B, buttress our initial findings reported in Section 1.4. There is a structural break in the wholesale prices which is most likely related to the 2005 merger. There are no effects of the upstream merger on consumer prices. The structural break in the wholesale prices, along with the absence of a corresponding structural break in the consumer prices, lends support to our theoretical model.

1.6 Conclusion

In this article we apply a theoretical model of a successive oligopoly with non-linear tariffs between upstream and downstream firms to make testable predictions of the possible anti-competitive effects of an upstream merger. The predictions are tested on an upstream merger in the Norwegian egg market. We find that the effects of this particular upstream merger are consistent with the predictions from the theoretical model. The upstream merger led to higher average prices to merged firm, while the prices to final consumers did not change.

The crucial element in the model is the non-linear contracts between the upstream and the downstream firms. With a non-linear contract, an increase in market power due to an upstream merger may affect mainly the fixed element and have no, or only a limited, effect on the price on the last unit. If that is the case, we can expect a small if any price effect on final consumers. Given that non-linear contracts between upstream and downstream firms are common in many industries, upstream mergers may often have quite limited or zero price effects on final consumers.

Ashenfelter and Hosken (2010) report the results of *ex post* evaluation of five US mergers. They consider those mergers that seem most likely to be problematic, and therefore present their results as an upper bound on the likely price effects on final consumers of mergers. However, all five mergers are upstream mergers and they investigate the price effect for final consumers. Our model and results indicate that non-linear contracts can dampen the pass-through rate. In that respect their study is probably not an upper bound on the price effect for final consumers following a horizontal merger between firms that sell directly to final consumers.

In 2010 the EU Commission cleared a merger between Unilever and Sara Lee, subject to structural remedies. Some brands owned by Sara Lee were sold. According to the Commission, this was “a strong and clear-cut remedy, [...] to ensure that the transaction would not lead to higher prices for consumers.”³² This was an upstream merger. Unilever and Sara Lee sold their products to retailers which, in turn, sold to final consumers. One important piece of evidence concerning price effects seems to be the results from a merger simulation model.³³ However, the vertical structure of the market is not taken into account in the merger simulation. The authors use scanner data on the consumer level to estimate demand, and this

³² See case COMP/M.5658 – Unilever/Sara Lee. The quote is from the press release from the EU Commission IP/10/1514 from November 17, 2010.

³³ See the Technical Annex to the decision, where the merger simulation model is described.

demand system is applied to estimate the price increase after the merger. This is the standard approach for merger simulations, but it nevertheless ignores the vertical dimension we are concerned about. In other words, the implicit assumption is full pass-through. As far as we understand it, contracts between upstream and downstream firms in this particular industry are not observed by rivals and they might include non-linear elements such as rebates and fixed fees. Our study suggests that in an industry with such a vertical structure there might be no price increase at all for the final consumers following an upstream merger. If so, the price increases estimated from the merger simulation model in this particular merger case may have been flawed.

The results of our analysis have important implications for merger assessment by competition authorities. Simply assuming that the price increase from an upstream merger will be passed on to final consumers can be misguided, as it apparently was in the upstream merger investigated in this article. The Norwegian Competition Authority's disapproval decision claimed – among other things – that it would lead to higher consumer prices. This turned out to be wrong. However, an upstream merger may have other effects not investigated here. For example, it may lead to less rivalry in R&D. This suggests that when assessing upstream mergers, competition authorities should be less concerned about the price effect on final consumers and more concerned about other effects, such as the effects on innovation.

Appendix A: Proofs

Proof of Proposition 1. The negotiating parties solve $\max_{w_{ir}, f_{ir}} N_{ir}$. The first order conditions are given by:

$$\begin{aligned} \frac{\partial N_{ir}}{\partial f_{ir}} &= 0 \Leftrightarrow \\ \gamma^{ir} (\pi_r - \nu_r^i) &= (1 - \gamma^{ir}) (\pi_i - \nu_i^r) \end{aligned}$$

and

$$\begin{aligned} \frac{\partial N_{ir}}{\partial w_{ir}} &= 0 \Leftrightarrow \\ \gamma^{ir} \frac{\frac{\partial \pi_i}{\partial w_{ir}}}{\pi_i - \nu_i^r} &= - (1 - \gamma^{ir}) \frac{\frac{\partial \pi_r}{\partial w_{ir}}}{\pi_r - \nu_r^i}. \end{aligned}$$

Combining the first order conditions yields:

$$\frac{\partial \pi_i}{\partial w_{ir}} + \frac{\partial \pi_r}{\partial w_{ir}} = 0.$$

Since rivals cannot respond to changes in w_{ir} , changes in w_{ir} affect the bilateral profit of producer i and retailer r only through the effects on the consumer prices at retailer r . Maximizing $(\pi_i + \pi_r) = Q_{ir}(\mathbf{w})(P_{ir}(\mathbf{w}) - w_{ir}) + \sum_{j=1, j \neq i}^N (Q_{jr}(\mathbf{w})(P_{jr}(\mathbf{w}) - w_{jr}) - f_{jr}) + Q_{ir}(\mathbf{w})(w_{ir} - c) + \sum_{s=1, s \neq r}^M (Q_{is}(\mathbf{w})(w_{is} - c) + f_{is})$ with respect to w_{ir} gives us a first order condition that can be written as:

$$\begin{aligned} &\left(\frac{\partial Q_{ir}}{\partial w_{ir}} (P_{ir}(\mathbf{w}) - w_{ir}) + Q_{ir}(\mathbf{w}) \frac{\partial P_{ir}}{\partial w_{ir}} \right) + \sum_{j=1, j \neq i}^N \left(\frac{\partial Q_{jr}}{\partial w_{ir}} (P_{jr}(\mathbf{w}) - w_{jr}) + Q_{jr}(\mathbf{w}) \frac{\partial P_{jr}}{\partial w_{ir}} \right) \\ &+ \frac{\partial Q_{ir}}{\partial w_{ir}} (w_{ir} - c) + \sum_{s=1, s \neq r}^M \frac{\partial Q_{is}}{\partial w_{ir}} (w_{is} - c) \\ &= 0. \end{aligned}$$

The first two terms of the left hand side can be written as

$$\begin{aligned} & \frac{\partial P_{ir}}{\partial w_{ir}} \left[\frac{\partial Q_{ir}}{\partial p_{ir}} (P_{ir}(\mathbf{w}) - w_{ir}) + Q_{ir} + \sum_{j=1, j \neq i}^N \frac{\partial Q_{jr}}{\partial p_{ir}} (P_{jr}(\mathbf{w}) - w_{jr}) \right] + \\ & \sum_{j=1, j \neq i}^N \frac{\partial P_{jr}}{\partial w_{ir}} \left[\frac{\partial Q_{jr}}{\partial p_{jr}} (P_{jr}(\mathbf{w}) - w_{jr}) + Q_{jr} + \frac{\partial Q_{ir}}{\partial p_{jr}} (P_{ir}(\mathbf{w}) - w_{ir}) + \sum_{k=1, k \neq i, j}^N \frac{\partial Q_{kr}}{\partial p_{jr}} (P_{kr}(\mathbf{w}) - w_{kr}) \right], \end{aligned}$$

and are zero by the first order conditions of the retailer in the retail price-setting stage. This leaves us with the following condition:

$$\frac{\partial Q_{ir}}{\partial w_{ir}} (w_{ir} - c) + \sum_{s=1, s \neq r}^M \frac{\partial Q_{is}}{\partial w_{ir}} (w_{is} - c) = 0. \quad (1.16)$$

Observe that if all other wholesale prices for product i are equal to marginal cost, this condition will be satisfied by also setting w_{ir} equal to marginal cost. We conclude that as long as the second order conditions are satisfied, there exists an equilibrium in which all wholesale prices are equal to marginal cost.

To see that this equilibrium is unique, note that for each producer i , we have a system of M first order conditions, each on the form of (1.16). Since own-price effects dominate cross price effects, the only way for all M conditions to hold simultaneously is that all wholesale prices are equal to marginal cost.

Derivation of Equation 1.7. The first order condition for f_{ir} can be written as:

$$\frac{\partial N_{ir}}{\partial f_{ir}} = 0 \Leftrightarrow$$

$$f_{ir} = \gamma^{ir} \left(\begin{aligned} & Q_{ir}(\mathbf{w}) (P_{ir}(\mathbf{w}) - w_{ir}) + \sum_{j=1, j \neq i}^N Q_{jr}(\mathbf{w}) (P_{jr}(\mathbf{w}) - w_{jr}) \\ & - \sum_{j=1, j \neq i}^N Q_{jr}(w_{ir} = \infty, \mathbf{w}_{-ir}) (P_{jr}(w_{ir} = \infty, \mathbf{w}_{-ir}) - w_{jr}) \end{aligned} \right)$$

$$- (1 - \gamma^{ir}) \left(\begin{aligned} & Q_{ir}(\mathbf{w}) (w_{ir} - c) + \sum_{s=1, s \neq r}^M Q_{is}(\mathbf{w}) (w_{is} - c) \\ & - \sum_{s=1, s \neq r}^M Q_{is}(w_{ir} = \infty, \mathbf{w}_{-ir}) (w_{is} - c) \end{aligned} \right)$$

In equilibrium, all marginal wholesale prices are equal to marginal cost, and the expression above is reduced to Equation 1.7.

Proof of Proposition 2. The parties of the negotiation solve $\max_{w_{1r}, w_{2r}, f_{ur}} N_{ur}$. The first order condition for product i ($i = 1, 2$) and f_{ur} are given by:

$$\frac{\partial N_{ur}}{\partial f_{ur}} = 0 \Leftrightarrow$$

$$\gamma^{ur} (\pi_r - \nu_r^u) = (1 - \gamma^{ur}) (\pi_u - \nu_u^r)$$

$$\frac{\partial N_{ur}}{\partial w_{ir}} = 0 \Leftrightarrow$$

$$\gamma^{ur} \frac{\frac{\partial \pi_u}{\partial w_{ir}}}{(\pi_u - \nu_u^r)} = - (1 - \gamma^{ur}) \frac{\frac{\partial \pi_r}{\partial w_{ir}}}{(\pi_r - \nu_r^u)}.$$

Combining the two gives us:

$$\frac{\partial \pi_u}{\partial w_{ir}} + \frac{\partial \pi_r}{\partial w_{ir}} = 0.$$

Consider now product 1. Differentiating $\pi_u + \pi_r$ with respect to w_{1r} and eliminating terms that are zero by the first order condition of the retailer leaves us with:

$$\sum_{i=1}^2 \left[\frac{\partial Q_{ir}}{\partial w_{1r}} (w_{ir} - c) + \sum_{s=1, s \neq r}^M \frac{\partial Q_{is}}{\partial w_{1r}} (w_{is} - c) \right] = 0. \quad (1.17)$$

If the wholesale prices from firm u to the other retailers are equal to marginal cost, the first order conditions for w_{1r} and w_{2r} will simultaneously be satisfied by setting $w_{1r} = w_{2r} = c$.

To see that this must be the case in any equilibrium, note that for firm u , we have a system of $2 * M$ first order conditions, each on the form of (1.17). To satisfy all first order conditions simultaneously, all wholesale prices must be equal to marginal cost.

The non-merging producers negotiate as before with the M retailers, taking the outcome of the other negotiations as given. Hence in equilibrium all marginal wholesale prices are unchanged and are equal to marginal cost.

Derivation of Equation 1.13.

$$\begin{aligned} \frac{\partial N_{ur}}{\partial f_{ur}} = 0 \Leftrightarrow \\ f_{ur} = \gamma^{ur} \left(\begin{aligned} & \sum_{i=1}^2 Q_{ir}(\mathbf{w}) (P_{ir}(\mathbf{w}) - w_{ir}) + \sum_{j=3}^N Q_{jr}(\mathbf{w}) (P_{jr}(\mathbf{w}) - w_{jr}) - \\ & \sum_{j=3}^N Q_{jr}(w_{1r} = w_{2r} = \infty, \mathbf{w}_{-1r,2r}) (P_{jr}(w_{1r} = w_{2r} = \infty, \mathbf{w}_{-1r,2r}) - w_{jr}) \end{aligned} \right) \\ - (1 - \gamma^{ur}) \left(\begin{aligned} & \sum_{i=1}^2 Q_{ir}(\mathbf{w}) (w_{ir} - c) + \sum_{s=1, s \neq r}^M \sum_{i=1}^2 Q_{is}(\mathbf{w}) (w_{is} - c) \\ & - \sum_{s=1, s \neq r}^M \sum_{i=1}^2 Q_{is}(w_{1r} = w_{2r} = \infty, \mathbf{w}_{-1r,2r}) (w_{is} - c) \end{aligned} \right) \end{aligned}$$

In equilibrium, all marginal wholesale prices are equal to marginal cost, and the expression above is reduced to Equation 1.13.

Appendix B: Robustness checks

As a robustness check, we apply the threshold procedure suggested by Hansen (2001) to estimate the timing of the break in wholesale prices of eggs. This amounts to splitting the sample under consideration into two subsamples, and test the equality of the parameters in the two subsamples using a Chow-statistic. Doing this for all potential breakdates, the break in the series is identified as where the Chow-statistics (one for each potential break date) are at their maximum. This value is called a Quandt-statistic. A high enough Quandt-statistic indicates that there is in fact a structural break in the time series. Critical values are provided in Andrews (1993).

Allowing for changes in the intercept and growth rate, the Quandt-statistic gives a strong indication of a break in the wholesale price process. The least squares estimator of the breakpoint is January 2005. A further test for two breakpoints in the wholesale egg prices is rejected. That the breakpoint is estimated to occur in January 2005 is not inconsistent with it being caused by the 2005 merger. That the merging parties were able and willing to coordinate their behavior a few months before the announcement of the merger, so-called gun jumping, seems perfectly plausible. When we use this breakpoint instead of the actual one, and estimate the models described in Section 1.4, we get results almost identical to the ones already reported in Table 1.1.

Using the Hansen procedure to find structural break for consumer egg prices, a Quandt-statistic is obtained which is below the critical values at a 5% significance level. There is, in other words, no indication of a structural break in the consumer price equation around the time of the merger. This lack of significant effects on the downstream firms' consumer prices is consistent with the model's theoretical prediction.

Bibliography

- Andrews, D. W. K. (1993). Tests for Parameter Instability and Structural Change With Unknown Change Point. *Econometrica* 61(4), pp. 821–856. DOI: 10.2307/2951764.
- Angrist, J. D. and J.-S. Pischke (2015). *Mastering 'metrics: The path from cause to effect*. Princeton University Press.
- Ashenfelter, O. and D. Hosken (2010). The effect of mergers on consumer prices: Evidence from five mergers on the enforcement margin. *Journal of Law and Economics* 53(3), pp. 417–466. DOI: 10.1086/605092.
- Ashenfelter, O., D. Hosken, and M. Weinberg (2013). The Price Effects of a Large Merger of Manufacturers: A Case Study of Maytag-Whirlpool. *American Economic Journal. Economic Policy* 5(1), pp. 239–261. DOI: 10.1257/pol.5.1.239.
- Bonnet, C. and P. Dubois (2010). Inference on vertical contracts between manufacturers and retailers allowing for nonlinear pricing and resale price maintenance. *RAND Journal of Economics* 41(1), pp. 139–164. DOI: 10.1111/j.1756-2171.2009.00093.x.
- Crooke, P. et al. (1999). Effects of assumed demand form on simulated postmerger equilibria. *Review of Industrial Organization* 15(3), pp. 205–217. DOI: 10.1023/A:1007730632277.
- Dobson, P. W. and M. Waterson (1997). Countervailing power and consumer prices. *The Economic Journal* 107(441), pp. 418–430. DOI: 10.1111/j.0013-0133.1997.167.x.
- Farrell, J. and C. Shapiro (1990). Horizontal mergers: an equilibrium analysis. *The American Economic Review* 80(1), pp. 107–126.
- (2010). Antitrust evaluation of horizontal mergers: An economic alternative to market definition. *The BE Journal of Theoretical Economics* 10(1). DOI: 10.2202/1935-1704.1563.

- Hansen, B. E. (2001). The new econometrics of structural change: Dating breaks in US labor productivity. *The Journal of Economic Perspectives* 15(4), pp. 117–128.
- Henrick Horn, A. W. (1988). Bilateral Monopolies and Incentives for Merger. *RAND Journal of Economics* 19(3), pp. 408–419.
- Inderst, R. and C. Wey (2003). Bargaining, Mergers, and Technology Choice in Bilaterally Oligopolistic Industries. *RAND Journal of Economics* 34(1), pp. 1–19.
- Manuszak, M. D. (2010). Predicting the impact of upstream mergers on downstream markets with an application to the retail gasoline industry. *International Journal of Industrial Organization* 28(1), pp. 99–111. DOI: 10.1016/j.ijindorg.2009.07.002.
- Milliou, C. and E. Petrakis (2007). Upstream horizontal mergers, vertical contracts, and bargaining. *International Journal of Industrial Organization* 25(5), pp. 963–987. DOI: 10.1016/j.ijindorg.2006.11.007.
- Norwegian Competition Authority (2005). *Hylleplassbetaling (Slotting Allowances)*. Report 2/2005.
- O’Brien, D. P. and G. Shaffer (2005). Bargaining, bundling, and clout: the portfolio effects of horizontal mergers. *RAND Journal of Economics* 36(3), pp. 573–595.
- Statens Landbruksforvaltning (2005). *Årsmelding 2004 (Annual Report 2004)*, Omsetningsrådet). Report.
- Ulsaker, S. A. (2013). *Konkurransanalyse i oppstrømsmarkeder (Competition assessment in upstream markets)*. SNF working paper 29/13.
- Ungern-Sternberg, T. von (1996). Countervailing power revisited. *International Journal of Industrial Organization* 14(4), pp. 507–519. DOI: 10.1016/0167-7187(95)00488-2.
- Villas-Boas, S. B. (2007a). Using retail data for upstream merger analysis. *Journal of Competition Law and Economics* 3(4), pp. 689–715. DOI: 10.1093/joclec/nhm030.
- (2007b). Vertical relationships between manufacturers and retailers: Inference with limited data. *The Review of Economic Studies* 74(2), pp. 625–652. DOI: 10.1111/j.1467-937X.2007.00433.x.
- Werden, G. J. (1996). A robust test for consumer welfare enhancing mergers among sellers of differentiated products. *The Journal of Industrial Economics* 44(4), pp. 409–413. DOI: 10.2307/2950522.

Chapter 2

Competing buyers, rent extraction and inefficient exclusion

SIMEN A. ULSAKER¹

Abstract

The article illustrates how a seller profitably can prevent entry of a potential competitor, even when entry would increase industry profit. Entry is prevented by offering exclusionary contracts to the buyers. The buyers are assumed to be differentiated firms, competing in a downstream market. Exclusion occurs in equilibrium as long as there is some degree of competition among the downstream firms, and even when there are no economies of scale in upstream production.

2.1 Introduction

Why would a buyer agree to sign an exclusive contract with a seller, if this precludes it from obtaining a valuable product from a potential entrant? The present article illustrates that when an established seller supplies two buyers that compete in a downstream market, the entry of an additional seller can reduce the joint profit of the buyers and the incumbent seller. When buyers compete, a buyer's flow profit

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gain from buying from the entrant is partly the other buyer's loss. The existence of these externalities among the buyers can enable the entrant to extract more profit from the industry than its contribution, something that would give the buyers and the incumbent a joint incentive to exclude the potential entrant. The article shows how and when the incumbent can profitably convince the buyers to sign exclusionary contracts and thereby exclude the entrant.

As pointed out by several authors, it is not obvious how an incumbent seller would find it profitable to sufficiently compensate its buyers to convince them not to trade with a more efficient seller.² A possible explanation is that economies of scale imply that the entrant will only enter if it is allowed to sell to several buyers. By exploiting coordination failures among the buyers or by making discriminatory offers, the incumbent may be able to convince a sufficient number of buyers to sign exclusionary contracts, even if the buyers would jointly be better off if entry occurred. Support for this explanation is given by Rasmusen et al. (1991) and Segal and Whinston (2000), who model an industry where the upstream firms offer identical products, and where the buyers are end-users (or firms operating in independent downstream markets).^{3,4}

Fumagalli and Motta (2006) were among the first to consider exclusionary contracts with competing buyers. They model an industry in which two identical buyers compete in prices in a downstream market. As in Rasmusen et al. (1991) and Segal and Whinston (2000), the upstream firms produce identical products. Fixed costs in the retail market make it prohibitively expensive for the incumbent to exclude the entrant, and entry occurs in equilibrium. The reason is that if one buyer has signed an exclusionary contract, the other has much to gain by refusing to do so, and being the only buyer free to trade with the more efficient entrant.

Abito and Wright (2008) qualify the findings of Fumagalli and Motta (2006).⁵

² Influential criticisms of the view that an incumbent can use exclusionary contracts to exclude a potential entrant to the detriment of its buyers is formulated by Posner (1976) and Bork (1978).

³ Aghion and Bolton (1987) consider a different motivation for the use of exclusionary contracts. In contrast to Rasmusen et al. (1991) and Segal and Whinston (2000), they allow contracts to be partially exclusive. This means that the transfer between the incumbent and a single buyer depends on whether or not the buyer trades with a potential entrant, which faces an unknown cost of production. By including an appropriate damages fee, paid by the buyer to the incumbent if and only if the buyer trades with the entrant, the two firms are able to extract some of the incremental profit generated by the entrant (when it enters).

⁴ Chen and Shaffer (2014) show how the use of appropriate minimum market share contracts can induce inefficient exclusion even in situations where exclusion does not occur when the incumbent demands exclusive dealing (that is when the minimum market share is equal to one).

⁵ In addition, Wright (2009) points out some errors in the analysis of Fumagalli and Motta (2006)

By removing the assumption of fixed costs of retailing, they illustrate how downstream competition can make exclusion a more, and not a less, likely outcome. Under exclusion, the incumbent can monopolize the downstream market by offering appropriate contracts to the downstream firms. When there is entry, competition between the upstream firms will drive the input prices down. It follows that when the downstream firms compete fiercely, total profit under entry can be very low. The incumbent may therefore be able to sufficiently compensate both buyers and exclude the entrant.⁶

Wright (2008) and Argenton (2010) consider the case where the upstream firms offer differentiated products. Wright (2008) models an industry where the upstream firms are horizontally differentiated, and where the downstream firms are identical and compete in prices. In this setting, exclusion can be profitable for similar reasons as the ones highlighted by Abito and Wright (2008): Exclusion leads to less competitive pricing and higher industry profit. Argenton (2010) studies a similar framework, but assumes that the upstream firms' products are vertically differentiated. He also finds that competition among the downstream firms facilitates exclusion. Under entry, the presence of the incumbent's inferior product would exercise competitive pressure, reducing the profit available through the sale of the entrant's product. As a result, the incumbent is able to sign exclusionary contracts with both downstream firms.

In contrast to what is common in the literature, the present article studies a setting where both upstream and downstream firms are differentiated. More specifically, I model an industry where two upstream firms offer differentiated products, and where the two buyers are differentiated competitors in a downstream market. A maintained assumption is that the fixed cost of entry is such that absent exclusionary contracts, the entrant would always enter and industry profit would increase. The incumbent is still able to profitably exclude the entrant without relying on miscoordination among the buyers or discriminatory exclusionary contracts.

In the baseline model, each buyer demands one unit of an input that comes in two types: a low-quality version produced by an incumbent and a high-quality version produced by an entrant. The profit of a downstream firm increases in own input quality, and decreases in the rival downstream firm's input quality. While industry

and shows that exclusion may in fact occur in equilibrium, if two-part tariffs are allowed.

⁶ As the authors note: "Since industry profit under monopoly is generally greater than under competition, downstream firms will often be willing to sign exclusive contracts to limit competition by preventing efficient entry" (Abito and Wright 2008, p. 228).

profit is higher when both downstream firms buy from the entrant, exclusion will be jointly profitable for the two downstream firms and the incumbent. This is not because exclusion allows them to dampen competition, but because, in the absence of exclusionary contracts, the entrant can exploit the fact that the buyers compete to extract higher payments from the downstream firms than what it “contributes” to the industry profit. The underlying intuition is simple. A downstream firm’s willingness to pay to buy the entrant’s product (rather than the incumbent’s) is the profit gain such a purchase would bring. Part of this profit gain is however the rival downstream firm’s loss.

Externalities among the buyers are what enable the entrant to extract more of the industry profit than its contribution. These externalities are also what enable the incumbent to profitably exclude the entrant without relying on discriminatory contracts or coordination failures among the buyers: Because the payments the entrant extracts exceed its contribution to the industry profit when both downstream firms are free to buy from it, the incumbent can offer exclusionary contracts that compensate both downstream firms according to what they can expect to earn under entry, and extract strictly more than this from the downstream firms after entry is prevented.

The baseline model is highly stylized. This allows us to derive precise predictions about equilibrium outcomes and to easily illustrate the underlying intuitions of the main results. The mechanisms behind the main results are however not driven by the particular restrictions of the baseline model. To illustrate this I also consider a more general setting where the upstream firms may be vertically or horizontally differentiated, and where each downstream firm may find it profitable to use strictly positive quantities (not necessarily equal to one) of the products of both upstream firms. The main results from the baseline model carry over to this setting. In any equilibrium in which both upstream firms sell to both downstream firms, the entrant extracts more of the industry profit than its contribution. Consequently, whenever the failure to sign exclusionary contracts with the downstream firms would bring about such an equilibrium, and the entry cost is such that the entrant will have to be free to sell to both downstream firms to cover its fixed entry costs, there will exist equilibria in which both downstream firms sign exclusionary contracts with the incumbent. In such exclusionary equilibria the joint payoff of the entrant and the downstream firms is strictly greater than in any entry equilibrium in which both downstream firms buy from both upstream firms. The underlying intuition is the

same as in the baseline model. A downstream firm's gain from buying from the entrant (and not only from the incumbent) is partly the rival downstream firm's (and the incumbent's) loss. The entrant thus extracts a higher share of the industry profit than its contribution, and the entrant can fully compensate both downstream firms and still gain from excluding the entrant.

The rest of the article is organized as follows. Section 2.2 introduces the framework of the baseline model. Some preliminary results are derived in Section 2.3. Section 2.4 examines the effect of letting the incumbent offer exclusionary contracts. Section 2.5 relaxes some of the assumptions of the baseline model to consider a more general setting. Section 2.6 concludes.

2.2 Baseline model

Two downstream firms, D_A and D_B , produce imperfectly substitutable products they sell on a downstream market. D_i ($i = A, B$) uses at most one unit of an indivisible input q_i in production. The input can be of two types: $q_i = 1$ signifies that the input used is of low quality, while $q_i = 2$ signifies that the input is of high quality ($q_i = 0$ signifies that the input is not used). The low-quality input is produced by an upstream firm, I (the incumbent). The high-quality input is produced by another upstream firm, E (the entrant). Let upstream production occur at a common and constant marginal cost, equal to c , and assume that the entrant has to incur a fixed entry cost $K \geq 0$ in order to be active in the market.

Denote the profit (gross of payments to upstream firms) of D_i by $\rho_i(q_i, q_j)$, where $q_i \in \{0, 1, 2\}$ specifies the input type used by D_i and $q_j \in \{0, 1, 2\}$ the input type used by the rival D_j ($j \neq i$). Assume that the downstream firms are symmetric, which enables us to write the profit function of D_i as $\rho(q_i, q_j)$. Further, assume that the profit generated at each downstream firm increases in own input quality and decreases in the input quality of the rival:

Assumption 1. $\rho(2, q_j) > \rho(1, q_j) > \rho(0, q_j) = 0$ for any q_j .

Assumption 2. $\rho(q_i, 0) > \rho(q_i, 1) > \rho(q_i, 2) > c$ for $q_i \in \{1, 2\}$.

Assume that industry profit is larger when both downstream firms use the high quality input than when neither uses it, that is, let $2\rho(2, 2) > \max\{2\rho(1, 1), \rho(1, 0)\}$. Further, assume that the gain from increasing own input quality is non-increasing in the quality of the rival.

Assumption 3. $\rho(\bar{q}_i, \underline{q}_j) - \rho(\underline{q}_i, \underline{q}_j) \geq \rho(\bar{q}_i, \bar{q}_j) - \rho(\underline{q}_i, \bar{q}_j)$ for all $(\bar{q}_i, \underline{q}_i, \bar{q}_j, \underline{q}_j)$ such that $\bar{q}_i > \underline{q}_i$ and $\bar{q}_j > \underline{q}_j$.

Assumption 3 is reminiscent of Assumption 3 in McAfee and Schwartz (1994), which states that a decrease in a firm's marginal costs is less valuable to it the lower a rival's marginal cost. In the present setting, the mechanism through which a higher input quality increases downstream profit is however left indeterminate. It may reduce the costs of downstream production, increase the quality of the downstream good, or both.

The preferences of the consumers are not modeled explicitly. I will however assume that the consumers benefit when the downstream firms use a higher quality input.

Assumption 4. *Consumer surplus is strictly increasing in q_i ($i = A, B$).*

The following assumption is made to assure the existence of (pure-strategy) equilibria in the case where exclusionary contracts are not allowed.

Assumption 5. $2(\rho(2, 2) - \rho(1, 2)) > \rho(2, 1) - \rho(1, 1)$.

A maintained assumption is that the fixed entry cost is such that the entrant will always decide to enter when exclusionary contracts are not allowed. As will become clear, this amounts to the following.

Assumption 6. $K \leq 2(\rho(2, 2) - \rho(1, 2))$.

2.3 Preliminary analysis

This section considers the situation where the incumbent is not able to write exclusionary contracts with the downstream firms. More formally, the interaction between the firms is modeled as follows. At t_1 , the entrant decides whether or not to incur the fixed cost K and enter the industry. At t_2 , active upstream firms offer publicly observable contracts to the downstream firms. The downstream firms then simultaneously reject or accept the received offers, with acceptance decisions becoming public information. At t_3 , downstream competition takes place and payments are made according to contracts.

It will be assumed that a contract between I (E) and D_i consists of a price $P_I \geq c$ ($P_E \geq c$) paid in exchange for the upstream firm's product.⁷ In order to limit the set of equilibria to consider, I do not allow the upstream firms to charge prices below cost. Following this restriction, we can without loss of generality normalize the marginal cost to zero. Observe that the upstream firms are not allowed to discriminate between the downstream firms when offering contracts. This is a common assumption in the literature on exclusive dealing. Assume that the entrant will enter if indifferent between entering and not. The solution concept used is that of pure strategy subgame-perfect Nash equilibrium (hereafter equilibrium).

The following lemma puts a lower bound on the gross payoff (*i.e.*, payoff gross of the fixed cost K) of the entrant, if it decides to enter.

Lemma 1. *When exclusionary contracts are not allowed and the entrant decides to enter, its gross payoff in any continuation equilibrium must be at least $2(\rho(2, 2) - \rho(1, 2))$.*

Proof. See appendix.

When the rival buys from the entrant, D_i gains $\rho(2, 2) - \rho(1, 2)$ in flow profit from buying from the entrant rather than from the incumbent. Lemma 1 establishes that the entrant is able to extract this amount from both downstream firms.

The following lemma characterizes the equilibrium outcome when exclusionary contracts are not allowed.

Lemma 2. *When exclusionary contracts are not allowed, both downstream firms will buy from the entrant in equilibrium. The payoff of the entrant will be $2(\rho(2, 2) - \rho(1, 2)) - K$, while each downstream firm gets a payoff equal to $\rho(1, 2)$. The incumbent gets a payoff equal to zero.*

Proof. See appendix.

An important observation follows from Lemma 2: The joint payoff of the incumbent and the downstream firms is always reduced by the presence of the entrant. Absent the entrant, the downstream firms would get a payoff equal to zero, but the incumbent would get at least $2\rho(1, 1)$, which is the industry profit when both

⁷ Free disposal is assumed. If D_i accepts a contract from the entrant, it will use the entrant's product. If D_i accepts a contract from the incumbent it will use this firm's product if and only if it does not accept a contract from the entrant.

downstream firms buy from the incumbent. In the presence of the entrant, the joint payoff of the downstream firms is $2\rho(1, 2)$, while the incumbent gets zero. The downstream firms' gain from entry is thus strictly less than the incumbent's loss, since $\rho(1, 1) > \rho(1, 2)$. Note that the joint payoff of the incumbent and the downstream firms is reduced by entry even though, since $2\rho(2, 2) > \max\{2\rho(1, 1), \rho(1, 0)\}$, entry increases the industry profit (gross of K).

The intuition is that the entrant is able to exploit externalities among the downstream firms. A downstream firm's willingness to pay to "upgrade" from the incumbent's to the entrant's input is the flow profit gain resulting from this upgrade. Part of this profit gain is however the rival downstream firm's loss. The existence of these externalities is what enables the entrant to extract more of the industry profit than its contribution to it. If the profit generated by one downstream firm were independent of the input used by the other (say if they operated in separate downstream markets), there would be no externalities, and the entrant would never be able to extract more profit than its contribution.

This observation invites the following question: Would the incumbent be able to use exclusionary contracts to exclude the entrant in a way that increased the joint payoff of itself and the two downstream firms, if such contracts were allowed? This question is taken up in the following section.

2.4 Exclusionary contracts

In this section we allow the incumbent a first mover advantage by adding a date t_0 before t_1 of the game specified in the previous section. At this date, the incumbent is allowed to offer the downstream firms publicly observable exclusionary contracts (more on these below).⁸ If accepted, an exclusionary contract makes it illegal for D_i to buy from the entrant. The incumbent will then be the only upstream firm offering D_i a contract at t_2 . If D_i rejects the offer at t_0 , both upstream firms are free to offer it a contract at t_2 . As before, the entrant is not allowed to discriminate between the buyers. The incumbent will be able to offer different prices to the downstream

⁸ If it was the firm with the superior product that had the first mover advantage, the outcome would be as in the previous section. As noted by Argenton (2010), the entrant's higher quality product may be the result of technical innovation. The incumbent's product can then be interpreted as an early version of the input, while the entrant can be thought of as a potential producer of a future and improved version. The identity of the entrant and the precise specification of its variety of the product may well be unknown when the incumbent is allowed to offer the downstream firms exclusionary contracts.

firms at t_2 if and only if one of them signed an exclusionary contract at t_0 and the other did not.

We know from the previous section that if both downstream firms reject the exclusionary contract offer, the entrant will decide to enter. Suppose now that one downstream firm has signed an exclusionary contract at t_0 (and is “signed”), while the other has not signed (and is “free”). The following lemma describes the continuation equilibrium when one downstream firm is free and the other downstream firm is signed, and the entrant decides to enter.

Lemma 3. *If one, and only one, downstream firm is free and the entrant enters, the entrant gets a payoff (gross of K) equal to $\rho(2, 1) - \rho(1, 1)$. The free downstream firm gets a payoff equal to $\rho(1, 1)$, while the signed downstream firm gets a payoff (gross of any upfront compensation received from the incumbent) equal to zero. The incumbent gets a payoff (gross of any upfront compensation paid to the signed downstream firm) equal to $\rho(1, 2)$.*

Proof. See appendix.

The signed downstream firm will always use the low quality input. Furthermore, the incumbent is willing to offer its product to the free downstream at cost. The entrant is therefore able to extract the free downstream firm’s gain from using the high quality input, namely $\rho(2, 1) - \rho(1, 1)$. We conclude that if only one downstream firm signs an exclusionary contract, the entrant will decide to enter if and only if $K \leq \rho(2, 1) - \rho(1, 1)$.

Simultaneous offers Let us first assume that the exclusionary contracts are offered simultaneously, and that they specify an upfront compensation x , paid by the incumbent to D_i if the exclusionary contract is signed. Furthermore, assume that the exclusionary contracts are non-discriminatory.⁹

Consider first the case where $K \leq \rho(2, 1) - \rho(1, 1)$. This means that the entrant will be able to cover its fixed cost K by selling only to one downstream firm. Consequently, the entrant will enter if at least one downstream firm is free. Then, there

⁹ The exclusionary contracts are assumed to be fully exclusionary, or “naked”. This means that that no transfer between D_i and the incumbent is allowed to depend on whether D_i trades with the entrant or whether the incumbent trades the rival downstream firm D_j . This restriction is common in the literature on exclusive dealing. A common justification is the belief that payments between two parties that are contingent on trade links including non-parties are difficult to enforce. See among others Segal and Whinston (2000) and Fumagalli and Motta (2006) for discussions.

will exist equilibria in which both downstream firms sign exclusionary contracts at t_0 and the entrant is excluded, but also equilibria in which one or both downstream firms are free and the entrant enters.

Proposition 1. *If $K \leq \rho(2, 1) - \rho(1, 1)$ and the incumbent simultaneously offers both downstream firms non-discriminatory exclusionary contracts, equilibrium outcomes are characterized as follows.*

Case A: $2\rho(1, 1) < \rho(1, 0)$. Both downstream firms sign exclusionary contracts at t_0 . Each downstream firm gets a payoff equal to $\rho(1, 1)$. The incumbent gets a payoff equal to $\rho(1, 0) - 2\rho(1, 1)$.

Case B: $2\rho(1, 1) \geq \rho(1, 0)$. There exists equilibria of the following types:

- (i) Both downstream firms sign exclusionary contracts at t_0 . Each downstream firm gets a payoff equal to $\rho(1, 1)$, and the payoff of both the entrant and the incumbent is zero.*
- (ii) Only one downstream firm signs an exclusionary contract at t_0 . The entrant sells to the free downstream firm and the incumbent to the signed downstream firm. The entrant gets a payoff equal to $\rho(2, 1) - \rho(1, 1) - K$. The payoff of the signed downstream firm is $\rho(1, 2)$, while the payoff of the free downstream firm is $\rho(1, 1)$. The payoff of the incumbent is zero.*
- (iii) Neither downstream firm signs an exclusionary contract at t_0 . The payoffs are as specified in Lemma 2.*

Proof. See appendix.

Proposition 1 establishes that exclusionary contracts may indeed be used to exclude the entrant, even when the fixed cost associated with entry is zero. In the case where $\rho(1, 0) > 2\rho(1, 1)$, that is, when industry profit is higher when only one downstream firm uses the low quality input than when both use it, the availability of exclusionary contracts not only leads to upstream exclusion, but also to downstream exclusion. In this case, the incumbent is able to get strictly positive profits by excluding the entrant, even when the entry cost K is zero. Not only the incumbent, but also the downstream firms gain from the use of exclusionary contracts: Their individual payoffs are strictly greater when exclusionary contracts are available than when they are not.

When $2\rho(1,1) \geq \rho(1,0)$, there still exists exclusionary equilibria. In this case, however, the incumbent has no real incentive to induce exclusion, as its equilibrium profit is zero regardless of whether exclusion occurs or not. Since a downstream firm that is alone in being free gets a payoff equal to $\rho(1,1)$, the entrant must offer the downstream firms a compensation equal to this amount to convince both to sign exclusionary contracts. Since $2\rho(1,1)$ equals the industry profit when the entrant is excluded, it follows that the incumbent is not able to exclude the entrant and earn a strictly positive payoff.

It is however worth noting that the downstream firms still gain from the use of exclusionary contracts, in the sense that their equilibrium payoffs are strictly greater when both sign an exclusionary contract than when neither does, and entry occurs. From the perspective of the incumbent and the downstream firms, equilibria where the entrant is excluded Pareto-dominate equilibria with entry. The entrant must thus rely on a coordination failure between the downstream firms and the incumbent in order to be able to enter. This is in a sense the opposite of what Rasmusen et al. (1991) and Segal and Whinston (2000) find when they consider the scope for exclusion when the incumbent simultaneously offers non-discriminatory contracts. In their setting, where economies of scale imply that the entrant needs more than one free buyer to be able to enter, the incumbent must rely on coordination failures among the buyers in order to profitably exclude the entrant.

Suppose now that $K > \rho(2,1) - \rho(1,1)$. This implies that the entrant will not be able to cover its fixed cost by only selling to one downstream firm, and that it will decide to enter only if both downstream firms are free. The incumbent will therefore only need to sign an exclusionary contract with one downstream firm to deter entry. The entrant is then always excluded, and the incumbent gets a strictly positive payoff in all equilibria.

Proposition 2. *If $\rho(2,1) - \rho(1,1) < F \leq 2(\rho(2,2) - \rho(1,2))$ and the incumbent simultaneously offers both downstream firms non-discriminatory exclusionary contracts, the entrant is excluded in all equilibria. The payoff of D_i is equal to $\Pi_i \in [0, \rho(1,2)]$. The incumbent gets a payoff equal to $\Pi_I \in [\max\{2\rho(1,1), \rho(1,0)\} - 2\rho(1,2), \max\{2\rho(1,1), \rho(1,0)\}]$.*

Proof. See appendix.

While Proposition 2 establishes that the entrant will be excluded in all equilibria, the equilibrium payoffs are somewhat indeterminate. The incumbent is always

able to extract $\max\{2\rho(1, 1), \rho(1, 0)\}$ from the downstream firms at t_2 . The question is how much it has to offer in compensation in order to convince at least one downstream firm to sign the exclusionary contract at t_0 . If a downstream firm believes that the rival will sign the exclusionary contract, it will be willing to sign in exchange for an arbitrarily low compensation, since it expects exclusion to occur regardless of its acceptance decision. If it believes that the rival will reject the offer, it will not be willing to accept for a compensation below $\rho(1, 2)$, since this is what it gets when both downstream firms are free. If we rule out equilibria that relies on miscoordination among the downstream firms at stage t_0 , that is, if we restrict attention to equilibria that are perfectly coalition-proof in the sense of Bernheim et al. (1987), the payoff of the incumbent will be exactly $\max\{2\rho(1, 1), \rho(1, 0)\} - 2\rho(1, 2)$ in all equilibria.¹⁰

Because the entrant extracts a higher payoff than its contribution to the industry profit when both downstream firms are free, the incumbent can compensate both downstream firms with what they can expect to get under entry upfront, and extract strictly more than this at t_2 . It is thus precisely the fact that the entrant extracts more from the industry than its contribution that enables the incumbent to profitably exclude the entrant, without offering discriminatory contracts or relying on coordination failures among the buyers. If the buyers were independent, say they operated in separate markets downstream, any perfectly coalition-proof equilibrium would give the incumbent a payoff equal to zero, regardless of whether exclusion occurred or not.

Regardless of the size of the entry cost, it will be in the joint interest of the downstream firms and the incumbent to exclude the entrant. The incumbent can therefore always afford to fully compensate both downstream firms with the amount they would get if both were free to buy from the entrant. What makes exclusion challenging to achieve when the entry cost is low (and the entrant only needs to sell to one downstream firm to enter), is that a downstream firm will get a relatively high payoff when it is alone in being free to buy from the entrant. If the rival downstream firm signs an exclusionary contract, it will be tempting for D_i to refuse to do so. A way for the incumbent to circumvent this problem is to offer *conditional* exclusionary contracts, whose exclusivity provision is binding only if both downstream firms

¹⁰ An equilibrium is perfectly coalition-proof if it continues to hold even when the downstream firms can make nonbinding agreements among themselves regarding their acceptance decisions at t_0 . Articles that use this concept include Segal and Whinston (2000) and Chen and Shaffer (2014).

accept the offer at t_0 .¹¹ With such contracts, both downstream firms will be free to buy from the entrant, unless both accept the exclusionary contract offered at t_0 . Then, D_i knows that if it rejects an offer at t_0 , both downstream firms will be free and it will get a payoff equal to $\rho(1, 2)$. Therefore, each downstream firm will be willing to sign an exclusionary contract for a compensation as small as $\rho(1, 2)$. Allowing such contracts, both downstream firms will sign an exclusionary contract at t_0 in all equilibria, regardless of the size of K .

Proposition 3. *If the incumbent simultaneously offers both downstream firms non-discriminatory exclusionary contracts in which the exclusivity provision is binding only if both buyers accept the offer, the entrant is excluded in equilibrium. The incumbent gets a payoff of equal to $\max\{2\rho(1, 1), \rho(1, 0)\} - 2\rho(1, 2)$. The payoff of D_i is equal to $\rho(1, 2)$.*

Proof. See appendix.

The ability to offer conditional exclusionary contracts facilitates exclusion because it makes it impossible for a downstream firm to be alone in being free to trade with the entrant. As a consequence, as long as the entrant offers each downstream firm a compensation equal to what it would make if both downstream firms were free, $\rho(1, 2)$, the downstream firms will not have an incentive (neither collectively or individually) to reject the offer. On the other hand, the incumbent cannot offer strictly less than $\rho(1, 2)$ in any equilibrium (regardless of the size of K), since a downstream firm always can secure itself this amount by rejecting the offer t_0 . Consequently, the equilibrium payoff of the incumbent is equal to $\max\{2\rho(1, 1), \rho(1, 0)\} - 2\rho(1, 2)$.

Sequential offers So far we have assumed that the incumbent is restricted to simultaneously offer the downstream firms non-discriminatory exclusionary contracts. Let us now instead assume that at t_0 , the incumbent first offers D_A an exclusionary contract. After the acceptance decision of D_A , the incumbent offers D_B a possibly different exclusionary contract, which is then accepted or rejected. The rest of the game then proceeds as before. The following proposition establishes that being able to offer contracts sequentially enables the incumbent to exclude the entrant and get a strictly positive payoff, regardless of the size of the entry cost K , and even when $2\rho(1, 1) \geq \rho(1, 0)$.

¹¹ I follow Wright (2009) in considering this type of exclusionary contract.

Proposition 4. *If $K \leq \rho(2, 1) - \rho(1, 1)$ and the incumbent is able to sequentially offer the downstream firms exclusionary contracts (D_A is offered a contract first), equilibrium outcomes are characterized as follows.*

Case A: $2\rho(1, 1) < \rho(1, 0)$. Both downstream firms sign exclusionary contracts. D_A gets a payoff $\Pi_A \in [\rho(1, 2), \rho(1, 1)]$, while D_B gets a payoff equal to $\rho(1, 1)$. The incumbent gets a payoff $\Pi_I \in [\rho(1, 0) - 2\rho(1, 1), \rho(1, 0) - \rho(1, 1) - \rho(1, 2)]$.

Case B: $2\rho(1, 1) \geq \rho(1, 0)$. There exists equilibria of the following types.

- (i) Both downstream firms sign exclusionary contracts. D_A gets a payoff $\Pi_A \in [\rho(1, 2), \rho(1, 1)]$, while D_B gets a payoff equal to $\rho(1, 1)$. The incumbent gets a payoff $\Pi_I \in [0, \rho(1, 1) - \rho(1, 2)]$.*
- (ii) Only D_B signs an exclusionary contract. The payoff of D_A is $\rho(1, 1)$ and the payoff of D_B is $\rho(1, 2)$. The payoff of the entrant is $\rho(2, 1) - \rho(1, 1) - K$, while the payoff of the incumbent is zero.*
- (iii) Neither downstream firm signs an exclusionary contract at t_0 . The payoffs are as specified in Lemma 2.*

Proof. See appendix.

Comparing Proposition 1 and Proposition 4, we see that the ability to offer contracts sequentially increases the upper bound on the incumbent's payoff in equilibria where both downstream firms sign exclusionary contracts.¹² Why does the ability to offer contracts sequentially help the incumbent? The reason is that if D_A rejects the exclusionary contract offer, there always exists continuation equilibria in which D_B does not sign an exclusionary contract and D_A earns $\rho(1, 2)$. This means that if D_A expects such a continuation equilibria to be played, it will be willing to sign an exclusionary contract for a compensation as small as $\rho(1, 2)$. If D_A signs, the incumbent will go on to sign up D_B in exchange for a compensation of $\rho(1, 1)$. Hence, the sum of the compensations necessary to induce exclusion may be strictly less than what the incumbent goes on to extract from the downstream firms at t_2 , namely $\max\{2\rho(1, 1), \rho(1, 0)\}$. Note also that even though the incumbent may now be able to extract a larger share of the industry profit when the entrant is excluded,

¹² Only having the ability to offer discriminatory contracts does not increase the upper bound on the equilibrium payoff of the incumbent for any K , but may make it cheaper to exclude the entrant without relying on miscoordination when the entrant only needs to sell to one downstream firm to enter.

exclusionary equilibria still Pareto dominate non-exclusionary equilibria from the perspective of the downstream firms.

Let us now consider the case where $K > \rho(2, 1) - \rho(1, 1)$, implying that the entrant will only enter if it is free to sell to both downstream firms. Then, allowing the incumbent to offer exclusionary contracts sequentially lets it exclude the entrant for free, without relying on miscoordination among the downstream firms.

Proposition 5. *If $\rho(2, 1) - \rho(1, 1) < F \leq 2(\rho(2, 2) - \rho(1, 2))$ and the incumbent is able to sequentially offer the downstream firms exclusionary contracts (D_A is offered a contract first), the entrant is excluded in all equilibria. Each downstream firm gets zero in payoff. The payoff of the incumbent is $\max\{2\rho(1, 1), \rho(1, 0)\}$.*

Proof. See appendix.

When the entrant needs both downstream firms to be free in order to profitably enter, D_A knows that if it refuses to sign an exclusionary contract, the incumbent will still exclude the entrant by signing an exclusionary contract with D_B . D_A will therefore be willing to sign an exclusionary contract without being compensated.

We have seen how the incumbent can use exclusionary contracts to exclude the entrant. As far as exclusionary contracts induce exclusion, they strictly reduce the quality of the input used by at least one downstream firm. Since consumer surplus is assumed to be increasing in the quality of the input used by the downstream firms, banning exclusionary contracts always (weakly) increases consumer surplus.

2.5 A more general framework

The model studied so far has been quite restrictive. Specifically, we have assumed that each downstream firm uses at most one unit of an (indivisible) input produced by the upstream firms. Furthermore, we have restricted attention to vertically differentiated upstream firms.

This section relaxes these assumptions and demonstrates that the mechanism behind the incentive to exclude the entrant, namely that the entrant is able to extract higher payments than its contribution to the industry profit, can also be present in less restrictive settings. To illustrate this, I now assume that the upstream firms offer imperfectly substitutable products to the downstream firms, and that the downstream firms resell these products in a downstream market, where they compete in prices. The upstream firms may be vertically or horizontally differentiated, and

each downstream firm may find it profitable to buy strictly positive quantities (not necessarily equal to one) of the products of both upstream firms.

As shown by Rey and Vergé (2010), in a setting like this, a “double common agency” equilibrium – that is, an equilibrium where both downstream firms buy from both upstream firms – often does not exist. When the downstream firms are competing in prices, allowing supply contracts to specify resale price (RPM) makes it easier to support double common agency equilibria. To avoid the problem of non-existence of equilibria, assume therefore that a supply contract between $h \in \{I, E\}$ and D_i consists of constant wholesale price w_{hi} , a fixed fee F_{hi} , and a price for the downstream market, p_{hi} .

Let $Q_{hi}(p)$ be the demand at downstream firm D_i for the product of upstream firm $h \in \{I, E\}$, when the prices in the downstream market are given by $p = (p_{IA}, p_{EA}; p_{IB}, p_{EB})$. Demand for either product at D_i is assumed to be increasing in the prices at the rival downstream firm. An implication of this is that demand for the incumbent’s product at a downstream firm that only sells this product increases if the rival downstream firm stops selling the entrant’s product:

$$\begin{aligned} Q_{IA}(p_{IA}, \infty; p_{IB}, \infty) &> Q_{IA}(p_{IA}, \infty; p_{IB}, p_{EB}), \\ \text{and} \\ Q_{IB}(p_{IA}, \infty; p_{IB}, \infty) &> Q_{IB}(p_{IA}, p_{EA}; p_{IB}, \infty), \end{aligned}$$

for any $p = (p_{IA}, p_{EA}; p_{IB}, p_{EB})$ such that demand is strictly positive for all four options.

Consider the case where the interaction between the firms is as follows. At t_0 , the incumbent, I , is able to offer the downstream firms exclusionary contracts. At t_1 , the entrant, E , decides whether or not to pay $K \geq 0$ and enter the market. At t_2 , active upstream firms offer supply contracts to the downstream firms. Acceptance decisions and downstream competition follow, and finally payments are made according to contracts.

Suppose now that both downstream firms are free, and that the entrant has decided to enter the market. Consider a possible continuation equilibrium in which both downstream firms buy from both upstream firms, with the price vector equal to $p = (p_{IA}, p_{EA}; p_{IB}, p_{EB})$. Then, as long as the continuation equilibrium after the contract offers at t_2 is unique, it must be the case that

$$F_{EA} = \sum_{h=I,E} (p_{hA} - w_{hA})Q_{hA}(p) - (p_{IA} - w_{IA})Q_{IA}(p_{IA}, \infty; p_{IB}, p_{EB}),$$

and

$$F_{EB} = \sum_{h=I,E} (p_{hB} - w_{hB})Q_{hB}(p) - (p_{IB} - w_{IB})Q_{IB}(p_{IA}, p_{EA}; p_{IB}, \infty).$$

That is, the fixed fee paid to the entrant by a downstream must equal the increase in flow profit resulting from accepting both contracts rather than only the contract from the incumbent. If the fixed fee exceeded this amount, the downstream firm would gain from rejecting the contract from the entrant, while if the fixed fee were lower, the entrant could slightly increase the fee and still get the downstream firm to accept the contract.

Consider now a possible double common agency equilibrium with $p = (p_{1I}, p_{1E}; p_{2I}, p_{2E})$, where $w_{Ii} \geq c_I$ and $p_{Ii} \geq w_{Ii}$ for $i = (A, B)$, and where at least one of these four inequalities holds strictly.¹³ Then, the joint payoff of the two downstream firms and the incumbent is given by

$$\begin{aligned} & \sum_{i=A,B} \sum_{h=I,E} (p_{hi} - w_{hi})Q_{hi}(p) - F_{Ei} \\ & + \sum_{i=A,B} (w_{Ii} - c_I)Q_{Ii}(p) \\ & = (p_{IA} - w_{IA})Q_{IA}(p_{IA}, \infty; p_{IB}, p_{EB}) + (p_{IB} - w_{IB})Q_{IB}(p_{IA}, p_{EA}; p_{IB}, \infty) \\ & + \sum_{i=A,B} (w_{Ii} - c_I)Q_{Ii}(p) \\ & < (p_{IA} - w_{IA})Q_{IA}(p_{IA}, \infty; p_{IB}, \infty) + (p_{IB} - w_{IB})Q_{IB}(p_{IA}, \infty; p_{IB}, \infty) \\ & + \sum_{i=A,B} (w_{Ii} - c_I)Q_{Ii}(p_{IA}, \infty; p_{IB}, \infty) \\ & = \sum_{i=A,B} (p_{Ii} - c_I)Q_{Ii}(p_{IA}, \infty; p_{IB}, \infty), \\ & \leq \Pi_I^m, \end{aligned}$$

¹³ Rey and Vergé (2010) study the case where the (horizontally differentiated) firms are symmetric both upstream and downstream, and look for symmetric equilibria in a linear demand example. While there in this setting exists equilibria in which the upstream margin is negative, these equilibria are Pareto-dominated from the upstream firms' perspective by equilibria with non-negative upstream margin.

where $\Pi_I^m \equiv \max_{(p_{IA}, p_{IB})} \sum_{i=A,B} (p_{Ii} - c_I) Q_{Ii}(p_{IA}, \infty; p_{IB}, \infty)$ is the maximal industry profit when the entrant is not active.

We observe that the equilibrium joint payoff of the downstream firms and the incumbent is strictly smaller than what they would be able to generate if the entrant were not in the market. We can conclude that in any equilibrium of the type considered above, the entrant is able to extract a higher share of the industry profit than its contribution to it. Consequently, whenever the downstream firms and the incumbent expect the entrant to enter and bring about a double common agency equilibrium, they will jointly have an incentive to exclude the entrant. The intuition is similar to the one in the baseline model: In order to buy from the entrant, a downstream firm is willing to pay the gain in flow profit from doing so (rather than just buying from the incumbent). However, since the downstream firms are competitors, some of this gain will be the rival downstream firm's (and the incumbent's) flow profit loss.

Assume now that the downstream firms are symmetric, and that following entry, double common agency continuation equilibria exist. Let $\bar{\pi}_E^{DCA}$ denote the entrant's payoff (gross of K) in the double common agency equilibrium that gives it the highest payoff. Then, as long as $K \leq \bar{\pi}_E^{DCA}$ and both downstream firms are free after t_0 , there will exist continuation equilibria in which the entrant enters, and both downstream firms buy from the entrant. While it then would be in the joint interest of the incumbent and the downstream firms to exclude the entrant at t_0 , it is not obvious that the incumbent could profitably convince the downstream firms to sign exclusionary contracts. As before, this would depend on the amount the downstream firms can expect to get if they reject the exclusionary offer at t_0 .

In contrast to what was the case in the baseline model, one can expect that there will exist multiple continuation equilibria both in the case where both downstream firms are free to buy from the entrant and in the case where only one downstream firm is free to do so. Whether or not the incumbent will be able to exclude the entrant may depend crucially on the payoffs the entrant and the free downstream firm expect to get if the entrant is only able to sell to one downstream firm. For example, if the entry cost K is strictly greater than what the entrant expects to get by only selling to one downstream firm, the incumbent only needs to sign an exclusionary contract with one downstream firm to prevent entry.

Consider now possible continuation equilibria when only one downstream firm is free to buy from the entrant. Let $\underline{\pi}_E^{PE}$ denote the payoff (gross of K) of the entrant

in the continuation equilibrium of this type that gives it the lowest payoff. Restricting attention to perfectly coalition-proof equilibria, we can state the following proposition.

Proposition 6. *Suppose that the incumbent simultaneously offers the downstream firms exclusionary contracts. Then, as long as $\underline{\pi}_E^{PE} < K \leq \bar{\pi}_E^{DCA}$, there exists (perfectly coalition-proof) equilibria in which the incumbent excludes the entrant by signing exclusionary contracts with both downstream firms.*

Proof. See appendix.

When $\underline{\pi}_E^{PE} < K \leq \bar{\pi}_E^{DCA}$, exclusion can be achieved in equilibrium by signing an exclusionary contract with one downstream firm. Since entry reduces the joint payoff of the incumbent and the downstream firms whenever both downstream firms buy from both upstream firms, the incumbent can offer both downstream firms a compensation equal what they expect to get under entry, and still increase its payoff. Whenever one free downstream firm is not enough to induce entry, the downstream firms will not have an individual (or joint) incentive to reject such an offer.

As was the case in the baseline model, if the entry cost is such that the entrant only needs to be able to sell to one downstream firm in order to enter the market, the ability of the incumbent to exclude the entrant may depend on the payoff a downstream firm is able to get by being the only downstream firm free to buy from the entrant. Consequently, it may be that the incumbent is unable to profitably exclude the entrant in equilibrium, even though entry reduces the joint payoff of the incumbent and the downstream firms. As before, the ability to offer conditional exclusionary contracts, whose exclusivity provision is binding only if both downstream firms accept the offer at t_0 , mitigates this problem and ensures the existence of exclusionary equilibria, regardless of the size of the entry cost K .

Proposition 7. *Suppose that the incumbent simultaneously offers both downstream firms non-discriminatory exclusionary contracts in which the exclusivity provision is binding only if both buyers accept the offer. Then, there exists (perfectly coalition-proof) equilibria in which the incumbent excludes the entrant by signing exclusionary contracts with both downstream firms, regardless of the size of K .*

Proof. See appendix.

When offered a conditional exclusionary contract, a downstream firm knows that both downstream firms will be free to buy from the entrant unless both sign

at t_0 . If the firms expect a specific symmetric double common agency equilibrium to be played whenever both downstream firms are free to buy from the entrant, the downstream firms will be willing to accept an exclusionary contract as long as it gives them a compensation at least equal to what they expect to get if both refuse to sign. As before, the incumbent will be able to compensate both downstream firms with this amount, and still gain strictly from excluding the entrant.

2.6 Conclusion

This article highlights how a potential entrant may be excluded in a setting that differs substantially from the settings normally studied by the theoretical literature on exclusionary contracts. The article illustrates how profitable exclusion can occur in equilibrium when differentiated upstream firms supply differentiated and competing downstream firms, even though entry would increase industry profit, and even when there are no fixed costs associated with entry. Exclusion occurs because the potential entrant would extract higher payments from the downstream firms than its contribution to the industry profit. The article complements earlier work, which have tended to focus on situations in which the buyers are either independent, or undifferentiated competitors. The article thus highlights that exclusionary contracts can be used to inefficiently, but profitably, exclude potential rivals in a much wider range of industries than what earlier work has suggested.

Appendix

This appendix contains proofs of all lemmas and propositions.

Proof of Lemma 1. Suppose otherwise, that is that E 's equilibrium gross payoff, denoted Π_E , is strictly below $2(\rho(2, 2) - \rho(1, 2))$.

Now, let E deviate by offering the downstream firms a contract with $P_E = \rho(2, 2) - \rho(1, 2) - \epsilon$, where $\epsilon \in (0, \frac{1}{2}[2(\rho(2, 2) - \rho(1, 2)) - \Pi_E])$. Following the deviation, the best response for D_i is to accept the offer from E , regardless of what its rival does. If the rival D_j accepts the offer from E , the payoff D_i can get by rejecting the offer from E is bounded from above by $\rho(1, 2)$, which is strictly below what it can get by accepting E 's offer, namely $\rho(1, 2) + \epsilon$. If D_j rejects the offer from E and accepts the offer from I , the payoff D_i can get by rejecting E 's offer is bounded from above by $\rho(1, 1)$. This is by Assumption 3 strictly less than what it can get by accepting the offer from E , namely $\rho(2, 1) - \rho(2, 2) + \rho(1, 2) + \epsilon$. If D_j rejects the offers from both upstream firms, the payoff D_i can get by rejecting E 's offer is bounded from above by $\rho(1, 0)$. This is less than what it can get by accepting E 's offer, namely $\rho(2, 0) - \rho(2, 2) + \rho(1, 2) + \epsilon$, again by Assumption 3.

Both downstream firm will thus accept E 's offer, leaving E with a gross payoff of $2(\rho(2, 2) - \rho(1, 2)) - 2\epsilon > \Pi_E$, which implies that the deviation is profitable.

Proof of Lemma 2. Note first that E must sell to at least one downstream firm in any equilibrium. If not, its gross payoff would be zero, contradicting Lemma 1. Furthermore, both downstream firms must be active in any equilibrium. Suppose otherwise, and let D_i be the inactive firm. Then both D_i and I will have a payoff equal to zero. If so, I could profitably deviate by offering its product to the downstream firms for price marginally above cost (which is equal to zero).

Suppose now that there exists an equilibrium in which I offer contracts with price P_I at t_2 , and where one downstream firm buys from E , while the other buys from I , contradicting the lemma. Let D_A be the firm that buys from E . Since D_B buys from I in equilibrium, it must be the case that this gives it a weakly positive payoff. It follows that $P_I \leq \rho(1, 2)$. Since D_A could get $\rho(1, 1) - P_I$ by only accepting I 's offer, E 's gross payoff is bounded from above by $\rho(2, 1) - \rho(1, 1) + P_I$.

Suppose now that E deviates by offering the downstream firms a contract with $P_E = \rho(2, 2) - \rho(1, 2) + P_I - \epsilon$, where $\epsilon \in (0, \frac{1}{2}[2(\rho(2, 2) - \rho(1, 2)) - \rho(2, 1) + \rho(1, 1) + P_I])$. Following this deviation, the best response for D_i is to accept the offer from E ,

regardless of what its rival does. If the rival D_j accepts the offer from E , the payoff D_i can get by rejecting the offer from E is given by $\rho(1, 2) - P_I \geq 0$, which is strictly below what it can get by accepting E 's offer, namely $\rho(1, 2) - P_I + \epsilon$. If D_j rejects the offer from E and accepts the offer from I , the payoff D_i can get by rejecting E 's offer is given by $\rho(1, 1) - P_I$. This is by Assumption 3 strictly less than what it could get by accepting the offer from E , namely $\rho(2, 1) - \rho(2, 2) + \rho(1, 2) - P_I + \epsilon$. If D_j rejects the offers from both upstream firms, the payoff D_i can get by rejecting E 's offer is given by $\rho(1, 0) - P_I$, which is strictly less than what it can get by accepting E 's offer, namely $\rho(2, 0) - \rho(2, 2) + \rho(1, 2) - P_I + \epsilon$, again by Assumption 3.

It follows that both downstream firms will accept the offer from E , which gives E a payoff equal to $2(\rho(2, 2) - \rho(1, 2)) + 2P_I - 2\epsilon$ following the deviation. This is strictly more than the upper bound on its equilibrium payoff, $\rho(2, 1) - \rho(1, 1) + P_I$. We conclude that the deviation is profitable. It follows that there cannot be an equilibrium in which only D_A buys from E . An analogous argument rules out equilibria in which only D_B buys from E , and we can conclude that both downstream firms will buy from E in any equilibrium.

Since both downstream firms buy from E , the equilibrium payoff of I is zero. It then follows that both downstream firms must get a payoff of at least $\rho(1, 2)$, since otherwise I could profitably deviate by offering its product for a price marginally above cost (zero). By Lemma 2, the payoff of E cannot be below $2(\rho(2, 2) - \rho(1, 2))$, which implies that the payoffs must be as specified in the lemma.

It is straightforward to verify that there exists an equilibrium in which I offers its product at a price equal to zero, E offers its product for a price equal to $\rho(2, 2) - \rho(1, 2)$, and both downstream firms accept the offer from E .

Proof of Lemma 3. Suppose that contrary to what is stated in the lemma, the continuation payoff of E after entry, denoted Π_E , is strictly below $\rho(2, 1) - \rho(1, 1)$. Now, let E deviate by offering a price $P_E = \rho(2, 1) - \rho(1, 1) - \epsilon$, where $\epsilon \in (0, (\rho(2, 1) - \rho(1, 1)) - \Pi_E)$. If the signed downstream firm buys from I , the free downstream firm gets at most $\rho(1, 1)$ by accepting the offer from I , which is strictly less than what it can get by accepting the offer from E , namely $\rho(1, 1) + \epsilon$. Furthermore, if the signed downstream firm does not buy from the incumbent, the free downstream firm can get at most $\rho(1, 0)$ by accepting the offer from I , which by Assumption 3 is strictly less than what it can get by accepting the offer from E , namely $\rho(2, 0) - \rho(2, 1) + \rho(1, 1) + \epsilon$.

Suppose now that contrary to what is stated in the lemma, the continuation payoff of the E , denoted Π_E , is strictly above $\rho(2, 1) - \rho(1, 1)$. This implies that the free downstream firm buys from E at a price P_E strictly above $\rho(2, 1) - \rho(1, 1)$. Furthermore, the continuation payoff of I (gross of compensation paid to the signed downstream firm) will be equal to $\rho(1, 2)$. Now, let I deviate by offering the free downstream firm a price $P_F = \epsilon_F$, where $\epsilon_F \in (0, P_E - (\rho(2, 1) - \rho(1, 1)))$ and the signed downstream firm a price $P_S = \rho(1, 2) - \epsilon_S$, where $\epsilon_S \in (0, \epsilon_F)$. The signed downstream firm will always accept its offer. It then follows that the free downstream firm also accepts the offer from I : It gets $\rho(1, 1) - \epsilon_F$ by doing so, which is strictly more than what it can get by accepting the offer from E , namely $\rho(2, 1) - P_E$. The payoff of I following the deviation is $\rho(1, 2) - \epsilon_S + \epsilon_F > \rho(1, 2)$, so we can conclude that the deviation is profitable.

Having established that the payoff of E is exactly $\rho(2, 1) - \rho(1, 1)$ in any continuation equilibrium after entry, it is plain to see that in any continuation equilibrium, the free downstream firm buys from the entrant, the signed from the incumbent, and that the payoffs are as specified in the lemma. Furthermore, it is straightforward to verify that there exists such a continuation equilibrium.

Proof of Proposition 1. Consider first Case A. If neither downstream firm signs an exclusionary contract with I , the equilibrium payoff of I is zero (from Lemma 2). This must also be the case in any equilibrium in which only one downstream firm signs an exclusionary contract with I , because by rejecting the exclusionary offer the signed downstream firm would get $\rho(1, 2)$ in any continuation equilibrium (again from Lemma 2). This implies that the compensation paid to this downstream firm as part of the exclusionary contract must be at least $\rho(1, 2)$, which is exactly what the incumbent is able to get by selling to the signed downstream firm at t_2 .

Now, assume that there exists an equilibrium in which at least one downstream firm rejects the exclusionary contract offer from I at t_0 . Then, as we have seen, the equilibrium payoff of I must be zero. Let I deviate by offering exclusionary contracts with compensation $x = \rho(1, 1) + \epsilon$, where $\epsilon \in (0, \frac{1}{2}[\rho(1, 0) - 2\rho(1, 1)])$. Both downstream firms will accept this offer, since the most any of them can get by rejecting the offer (and bringing about the entry of E) is $\rho(1, 1)$. After both have signed, I will sell to only one downstream firm at t_0 , and extract $\rho(1, 0)$ from this firm. The payoff of I following the deviation is therefore $\rho(1, 0) - 2\rho(1, 1) - 2\epsilon > 0$. We therefore conclude that the deviation is profitable, and that both downstream

firms must sign exclusionary contracts with I in any equilibrium.

When both downstream firms sign exclusionary contracts, they get their entire payoff through the compensation x . This compensation cannot be strictly greater than $\rho(1, 1)$ in equilibrium since I then could slightly decrease the compensation and still get both downstream firms to sign. The compensation cannot be strictly less than $\rho(1, 1)$ in equilibrium either, since this is what a downstream firm will get by rejecting the offer at t_0 (when the rival accepts). We therefore conclude that the equilibrium payoffs must be as specified in the proposition. It is straightforward to verify that there indeed exists an equilibrium in which I offers a compensation of $\rho(1, 1)$ and both downstream firms accept.

Consider now Case B. Consider first a possible equilibrium in which both downstream firms signs exclusionary contracts at t_0 . Then the downstream firms get their entire payoffs through the compensation x . Either downstream firm could get a payoff equal to $\rho(1, 1)$ by rejecting the exclusionary contract and thereby inducing entry. This means that the compensation x must be at least $\rho(1, 1)$ in any such equilibrium. Consequently, I would have to pay $2\rho(1, 1)$ in compensation in such an equilibrium, which is exactly what it can extract at t_0 when both downstream firms are signed. It follows that the payoffs in such an equilibrium must be as specified in the proposition. It is straightforward to verify that there exists an equilibrium of type (i) in which I offers a compensation of $\rho(1, 1)$ and both downstream firms accept.

Suppose now that there exists an equilibrium in which only one downstream firm signs the exclusionary contract. Since this firm could $\rho(1, 2)$ by rejecting the exclusionary offer and being free to buy from the entrant at t_2 , the compensation x must be at least $\rho(1, 2)$. It cannot be strictly more, since $\rho(1, 2)$ is what I can extract from this downstream firm at t_2 . It follows that the payoffs in any equilibrium of this type must be as specified in the proposition. It is straightforward to verify that there indeed exists an equilibrium of type (ii) in which I offer a compensation of $\rho(1, 2)$ and only one downstream firm accepts.

The reasoning above implies that the payoff of I must be zero in any continuation equilibrium in which at least one downstream firm signs an exclusionary contract at t_0 . It is therefore straightforward to see that there also exists equilibria of type (iii) in which I offer unacceptable exclusionary contracts at t_0 , gets zero in payoff and makes both downstream firms free to buy from E . We then know from the analysis in Section 2.3 that the equilibrium payoffs must be as specified in Lemma 2.

Proof of Proposition 2. Suppose first that contrary to what is stated in the proposition, I 's equilibrium payoff, denoted Π_I , is strictly below $\max\{2\rho(1, 1), \rho(1, 0)\} - 2\rho(1, 2)$. Now let I deviate by offering exclusionary contracts with $x = \rho(1, 2) + \epsilon$, where $\epsilon \in (0, \frac{1}{2}[\max\{2\rho(1, 1), \rho(1, 0)\} - 2\rho(1, 2) - \Pi_I])$. Regardless of the acceptance decision of its rival D_j , the best response of D_i is to accept this offer. If the rival D_j accepts the offer, D_i would get a payoff equal to zero if it rejected the offer, whereas it gets $\rho(1, 2) + \epsilon$ by accepting the offer. If D_j rejects the offer, D_i would get $\rho(1, 2)$ by also rejecting the offer and bringing about the entry of E , which is strictly below what it could get by accepting the offer, $\rho(1, 2) + \epsilon$. It follows that both downstream firms must accept the offer in any continuation equilibrium. The payoff of I following the deviation is then $\max\{2\rho(1, 1), \rho(1, 0)\} - 2\rho(1, 2) - 2\epsilon > \Pi_I$, which implies that the deviation is profitable.

Since I would get zero in any equilibrium in which both downstream firms are free, it follows that E must be excluded in any equilibrium. The compensation a downstream firm will demand in order to sign an exclusionary contract depends on whether it believes its rival to accept. If it expects its rival to accept, it would not have an incentive to reject even when offered a compensation equal to zero, while if it expects its rival to reject, it would never accept for a compensation strictly below $\rho(1, 2)$. It is straightforward to verify that there exists equilibria for any compensation in the range $[0, \rho(1, 2)]$, implying that the equilibrium payoffs will be as specified in the proposition.

Proof of Proposition 3. With conditional exclusionary contracts, each downstream firm will get a payoff equal to $\rho(1, 2)$ by refusing to sign the exclusionary contract (because then both will be free and the entrant will enter), and a payoff equal to the upfront compensation x if both sign the exclusionary contract.

Suppose that contrary to what is stated in the proposition, E enters in equilibrium. Then we know that the equilibrium profit of I is zero. Now, let I deviate by offering conditional exclusionary contracts with compensation $x = \rho(1, 2) + \epsilon$, where $\epsilon \in (0, \frac{1}{2}[\max\{2\rho(1, 1), \rho(1, 0)\}])$. Since a downstream firm would get $\rho(1, 2)$ if it rejected this offer, it follows that both downstream firms will accept the offer. The profit of I will therefore be $\max\{2\rho(1, 1), \rho(1, 0)\} - 2\epsilon > 0$, implying that the deviation is profitable. We conclude that E is excluded in any equilibrium.

Since a downstream firm gets $\rho(1, 2)$ by rejecting an exclusionary contract offer regardless of the acceptance decision of its rival, it is straightforward to verify that

in the unique equilibrium I offers exclusionary contracts with $x = \rho(1, 2)$ and both downstream firms accepts. The payoffs will then be as specified in the proposition.

Proof of Proposition 4. Consider a possible equilibrium in which D_A rejects the exclusionary contract offer. Then we know from Lemma 2 and Lemma 3 that E will enter and sell to D_A . If D_B also rejects its exclusionary contract offer, the payoff of I will be zero. Since D_B can always secure itself $\rho(1, 2)$ by rejecting the exclusionary contract, it follows that it must get at least this amount in compensation in any continuation equilibrium in which it signs an exclusionary contract. Since the most I can extract from D_B at t_2 is $\rho(1, 2)$, it then follows that the continuation equilibrium payoff of I is always zero if D_A rejects. Therefore, following a rejection by D_A there exists continuation equilibria in which D_B signs an exclusionary contract in exchange for a compensation equal to $\rho(1, 2)$, and continuation equilibria in which D_B does not sign an exclusionary contract.

Let us now suppose that D_A accepts the exclusionary contract offer. Then, D_B will be willing to sign an exclusionary contract as long as the compensation x_B is at least $\rho(1, 1)$, since this is what it can get by rejecting the exclusionary contract offer. By making D_B sign an exclusionary contract, I can guarantee itself a continuation payoff (gross of x_A and x_B) equal to $\max\{2\rho(1, 1), \rho(1, 0)\}$, while its continuation payoff will be $\rho(1, 2)$ if it fails to sign an exclusionary contract with D_B . Since I 's gain from signing up D_B , $\max\{2\rho(1, 1), \rho(1, 0)\} - \rho(1, 2)$, is strictly greater than what D_B must get to be willing to sign an exclusionary contract, that is $\rho(1, 1)$, it follows that if D_A signs an exclusionary contract, D_B will also sign one in any continuation equilibrium. Furthermore, it must be the case that D_B signs in exchange for a compensation exactly equal to $\rho(1, 1)$, since otherwise I could always slightly reduce the compensation and still get D_B to sign. We conclude that by convincing D_A to sign, I secures itself a continuation payoff of $\max\{2\rho(1, 1), \rho(1, 0)\} - \rho(1, 1)$, gross of the compensation paid to D_A .

Consider now Case A. Suppose that there exists an equilibrium in which D_A rejects the exclusionary contract offer. Then the equilibrium payoff of I must be zero. Now let I deviate by offering D_A an exclusionary contract with a compensation x_A equal to $\rho(1, 1) + \epsilon$, where $\epsilon \in (0, \rho(1, 0) - 2\rho(1, 1))$. Since $\rho(1, 1)$ is the most D_A can get by rejecting this offer, the offer will be accepted. Following this, D_B will sign an exclusionary contract in exchange for a compensation equal to $\rho(1, 1)$. It follows that the payoff of I following the deviation is $\rho(1, 0) - 2\rho(1, 1) - \epsilon > 0$,

implying that the deviation is profitable.

We conclude that D_A (and hence also D_B) must sign an exclusionary contract in any equilibrium. We have already established that D_B must get a compensation (and payoff) equal to $\rho(1, 1)$ in any such equilibrium. How much D_A must get in order to accept an exclusionary contract depends on what it can expect to get by rejecting the offer at t_0 . As noted above, there exists continuation equilibria following a rejection by D_A in which D_B goes on to sign an exclusionary contract. In these continuation equilibria the payoff of D_A is $\rho(1, 1)$. There also exist continuation equilibria in which D_B does not sign an exclusionary contract and the payoff of D_A is $\rho(1, 2)$. Subgame perfection demands not only that the players expect Nash equilibria to be played in every subgame, but also that the players agree on which equilibrium that will be played in a given subgame. However, since different offers of x_A at t_0 give rise to different subgames, the payoff D_A can expect by rejecting the offer at t_0 may depend on x_A . Therefore, there exists equilibria for any compensation x_A in the range $[\rho(1, 2), \rho(1, 1)]$. (If any rejection leads to a continuation equilibrium in which D_B does not sign an exclusionary contract, D_A could be willing to sign for as little as $\rho(1, 2)$ in compensation. If any rejection leads to a continuation equilibrium in which D_B signs an exclusionary contract D_A will not sign for a compensation below $\rho(1, 1)$.) Consequently, the equilibrium payoffs will be as specified in the proposition.

Let us now turn to Case B. We know that if D_A signs an exclusionary contract, D_B will also sign one in any continuation equilibrium. As in Case A, the payoff D_A can expect to get by rejecting the offer at t_0 may depend on x_A . Therefore, there will exist exclusionary equilibria of type (i) for any compensation x_A in the range $[\rho(1, 2), \rho(1, 1)]$ in which D_A signs an exclusionary contract at t_0 in exchange for x_A .

There also exists equilibria in which E enters. For example, if rejection following any offered compensation $x_A \in [\rho(1, 2), \rho(1, 1)]$ will lead to a continuation equilibrium where D_B signs an exclusionary contract, D_A will never sign for a compensation strictly below $\rho(1, 1)$. Then I must compensate the buyers with at least as much as it can expect to extract at t_2 in order to induce exclusion, and it follows that there exists equilibria of both type (ii) and (iii).

Proof of Proposition 5. If D_A rejects the first offer, D_B will still sign an exclusionary contract in any continuation equilibrium. To see this, note that D_B would always sign for any compensation strictly above $\rho(1, 2)$, whereas I would be willing to offer

a compensation up to $\max\{2\rho(1,1), \rho(1,0)\}$. Understanding this, D_A would accept to sign in exchange for any strictly positive compensation. It follows that in any equilibrium D_A will sign an exclusionary contract in exchange for a compensation of zero. A strictly positive compensation cannot be offered in equilibrium, because I could then always slightly reduce the compensation and still induce D_A to sign. The entrant will then be excluded, and the incumbent will extract the entire industry profit at t_2 . Whether or not D_B signs an exclusionary contract (in exchange for a compensation equal to zero) does not affect the equilibrium payoffs.

Proof of Proposition 6. Suppose that if only one downstream firm is free, E gets a continuation payoff (gross of K) equal to π_E^{PE} in all subgames following entry. Then it will be sufficient for the incumbent to sign up one downstream firm at t_0 to exclude the entrant. By excluding the entrant, the incumbent can extract Π_I^m from the downstream firms at t_2 . Remember that Π_I^m is strictly greater than the joint payoff of the incumbent and the downstream firms in any double common agency equilibrium. Then, if the players expect a given (symmetric) double common agency continuation equilibrium to be played whenever both downstream firms are free, there exists a perfectly coalition-proof equilibrium in which the incumbent offers (and the downstream firms accept) an exclusionary contract where each downstream firm gets a compensation equal to what it expects to get when both downstream firm reject the exclusionary contract offer, leaving the incumbent with a strictly higher payoff than what it would get under entry.

If one downstream firm rejected this offer, the entrant would still be excluded, giving the deviating downstream firm a payoff equal to zero. This, neither downstream firm has an individual incentive to deviate. A joint deviation where both downstream firms reject the offer is also not profitable, since then both downstream firms would get a payoff equal to the compensation they get by accepting the offer from the incumbent. Since we are considering perfectly coalition-proof equilibria, the incumbent cannot have an incentive to deviate and offer a lower compensation at t_0 , since the downstream firms would then have a joint incentive to both reject the exclusionary contract and bring about entry. We conclude that there does indeed exist a perfectly coalition-proof exclusionary equilibrium as specified in the proposition.

Proof of Proposition 7. By excluding the entrant the incumbent can extract Π_I^m from the downstream firms at t_2 , which is strictly greater than the joint payoff of

the incumbent and the downstream firms in any double common agency equilibrium. Then, if the players expect a given (symmetric) double common agency continuation equilibrium to be played whenever both downstream firms are free, there exists a perfectly coalition-proof equilibrium in which the incumbent offers (and the downstream firms accept) an exclusionary contract where each downstream firm gets a compensation equal to what it expects to get when both downstream firm reject the exclusionary contract offer, leaving the incumbent with a strictly higher payoff than what it would get under entry.

If one downstream firm rejected this offer, both downstream firms would be free to buy from the entrant and the deviating downstream firm would get a payoff equal to what it gets by accepting the offer. A joint deviation where both downstream firms reject the offer is also not profitable, since then both downstream firms would get a payoff equal to the compensation they get by accepting the offer from the incumbent. The incumbent has no incentive to deviate and offer a lower compensation at t_0 , since each downstream firm would have an incentive to reject the exclusionary contract and bring about entry. We conclude that there does indeed exist a perfectly coalition-proof exclusionary equilibrium as specified in the proposition.

Bibliography

- Abito, J. M. and J. Wright (2008). Exclusive dealing with imperfect downstream competition. *International Journal of Industrial Organization* 26(1), pp. 227–246. DOI: 10.1016/j.ijindorg.2006.11.004.
- Aghion, P. and P. Bolton (1987). Contracts as a Barrier to Entry. *The American Economic Review* 77(3), pp. 388–401.
- Argenton, C. (2010). Exclusive quality. *The Journal of Industrial Economics* 58(3), pp. 690–716. DOI: 10.1111/j.1467-6451.2010.00424.x.
- Bernheim, B. D., B. Peleg, and M. D. Whinston (1987). Coalition-Proof Nash Equilibria I. Concepts. *Journal of Economic Theory* 42(1), pp. 1–12. DOI: [http://dx.doi.org/10.1016/0022-0531\(87\)90099-8](http://dx.doi.org/10.1016/0022-0531(87)90099-8).
- Bork, R. H. (1978). *The Antitrust Paradox*. New York: Basic Books.
- Chen, Z. and G. Shaffer (2014). Naked exclusion with minimum-share requirements. *RAND Journal of Economics* 45(1), pp. 64–91. DOI: 10.1111/1756-2171.12042.
- Fumagalli, C. and M. Motta (2006). Exclusive Dealing and Entry, when Buyers Compete. *American Economic Review* 96(3), pp. 785–795. DOI: 10.1257/aer.96.3.785.
- McAfee, R. P. and M. Schwartz (1994). Opportunism in Multilateral Vertical Contracting: Nondiscrimination, Exclusivity, and Uniformity. *The American Economic Review* 84(1), pp. 210–230.
- Posner, R. A. (1976). *Antitrust Law: An Economic Perspective*. University of Chicago Press.
- Rasmusen, E. B., J. M. Ramseyer, and J. S. Wiley (1991). Naked Exclusion. *American Economic Review* 81(5), pp. 1137–45.
- Rey, P. and T. Vergé (2010). Resale price maintenance and interlocking relationships. *The Journal of Industrial Economics* 58(4), pp. 928–961. DOI: 10.1111/j.1467-6451.2010.00439.x.

- Segal, I. and M. D. Whinston (2000). Naked Exclusion: Comment. *American Economic Review* 90(1), pp. 296–309. DOI: 10.1257/aer.90.1.296.
- Wright, J. (2008). Naked exclusion and the anticompetitive accommodation of entry. *Economics Letters* 98(1), pp. 107–112. DOI: 0.1016/j.econlet.2007.04.014.
- (2009). Exclusive dealing and entry, when buyers compete: Comment. *The American Economic Review* 99(3), pp. 1070–1081. DOI: 10.1257/aer.99.3.1070.

Chapter 3

Interlocking relationships and quantity rationing

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Abstract

As shown by Rey and Vergé (2010), resale price maintenance can be used to maximize industry profit when competing manufacturers sell their brands through the same competing retailers. The present article shows how quantity rationing non-linear tariffs that place no explicit restrictions on the behavior of the retailers in the downstream market can be used to achieve the same outcome.

3.1 Introduction

Upstream manufacturers often distribute their products through several competing downstream firms, and downstream firms often distribute the products of several competing manufacturers. To name two of many examples, competing grocery chains often stock the same brands, and the same television channels are often available through competing distributors. The present article studies the use of quantity rationing tariffs in such a setting, and illustrates how they can be used to dampen competition and support the industry profit maximizing outcome.

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Rey and Vergé (2010) model an industry with “interlocking relationships”: Two manufacturers have the opportunity of selling their brands through the same two competing retailers. They find that when the manufacturers make public take-it-or-leave-it offers to the retailers, and the retailers must reject both or accept both offers, simple two-part tariffs are not sufficient to keep retail prices high enough to maximize industry profit. Resale price maintenance (RPM) does however suffice to completely eliminate competition at both levels in the vertical chain, and raise retail prices to the level that maximizes industry profit.

The present article illustrates that in this setting, the anticompetitive effect of RPM is a result of the assumption that the manufacturers offer two-part tariffs. If allowed to offer general non-linear tariffs, the manufacturers can induce an equilibrium in which industry profit is maximized, without leaving the retailers any rent. In order to achieve this outcome, it is sufficient to use quantity rationing tariffs, where the payment makes a jump if a retailer chooses to buy more than a given quantity from a manufacturer.

3.2 Framework

Consider an industry where two manufacturers, manufacturer A and manufacturer B , sell differentiated brands through two retailers, retailer 1 and retailer 2. Manufacturer i ($i = A, B$) incurs a production cost $C_i(q_{i1}, q_{i2}) = c_i(q_{i1} + q_{i2})$, where q_{ij} is the quantity retailer j ($j = 1, 2$) buys from the manufacturer, and $c_i \geq 0$ is the constant marginal cost of production. The retailers incur no other costs than what they pay the manufacturers.

Rey and Vergé (2010) consider price competition at the retail level. I here follow Marx and Shaffer (2007) and Rey and Whinston (2013) in remaining agnostic about the exact form of downstream competition. Let $q \equiv (q_{A1}, q_{B1}; q_{A2}, q_{B2})$ be the vector of quantities sold by the retailers in the downstream market. Further, let $\rho_{ij}(q) \equiv \rho_{ij}(q_{ij}, q_{hj}; q_{ik}, q_{hk})$ be the revenue generated by the sale of any brand $i \neq h \in \{A, B\}$ at any retailer $j \neq k \in \{1, 2\}$. The end-consumers view the retailers as substitutes. Assume therefore that $\rho_{ij}(\cdot)$ is strictly decreasing in the rival retailer’s quantities, q_{ik} and q_{hk} . Denote total revenue by $\rho(q) \equiv \rho_{A1}(q) + \rho_{B1}(q) + \rho_{A2}(q) + \rho_{B2}(q)$, and denote the industry profit by $\Pi(q) \equiv \rho(q) - c_A(q_{A1} + q_{A2}) - c_B(q_{B1} + q_{B2})$. Assume that there is a quantity vector $q^M = (q_{A1}^M, q_{B1}^M; q_{A2}^M, q_{B2}^M)$ that uniquely maximizes the industry profit, and denote the maximized profit as Π^M . Finally, let $\Pi_i(q) \equiv$

$\rho_{i1}(q) + \rho_{i2}(q) - c_i(q_{i1} + q_{i2})$ denote the profit generated by the sale of brand i .

While general contracts are allowed, special attention will be given to tariffs in which the mapping from quantity ordered, q_{ij} , to payment, T_{ij} , takes the following form:

$$T_{ij}(q_{ij}) = \begin{cases} F_{ij} & \text{if } q_{ij} \leq \bar{q}_{ij} \\ \infty & \text{if } q_{ij} > \bar{q}_{ij}. \end{cases}$$

A tariff of this form implies that by paying a fixed fee F_{ij} , retailer j can buy manufacturer i 's brand without additional cost, up until the threshold quantity $\bar{q}_{ij}$. Buying strictly more than this quantity leads to an infinitely high payment. For simplicity, denote a tariff of this sort by $(F_{ij}, \bar{q}_{ij})$. Such a tariff is in effect equivalent to the point-contracts studied by Rey and Whinston (2013), as long as we assume that retailers can freely dispose unsold quantities.

3.3 Intrinsic double common agency

This article considers an intrinsic double common agency situation: The market is assumed to break down if a retailer refuses to carry a brand. While this assumption will be violated in most real-world markets, the discussion in the next section illustrates that the main result of this section can be expected to carry over to more realistic market situations. Formally, consider the following game.

Stage 1 Each manufacturer i ($i = A, B$) offers a contract to each retailer j ($j = 1, 2$). Contract offers are simultaneous and public.

Stage 2 The retailers simultaneously decide whether to reject or accept the offers. Acceptance decisions are public information. If all offers are accepted, the game proceeds to Stage 3. If at least one offer is rejected, the game ends, and all firms earn zero in profit.

Stage 3 Downstream competition occurs, and payments are made according to contracts.

The solution concept used in this article is subgame-perfect Nash equilibrium (hereafter equilibrium). General contracts are allowed, but a contract is not allowed to include explicit restrictions on the behavior of the retailer in the downstream market.

The main result of this article is the following.

Proposition 1. *There exists an equilibrium in which manufacturer i offers retailer j the contract $(q_{ij}^M, \rho_{ij}(q^M))$, and both retailers accept both contract offers. In this equilibrium the industry profit is maximized and both retailers earn a profit of zero. The equilibrium profit of manufacturer i is $\Pi_i(q^M) \equiv \rho_{i1}(q^M) + \rho_{i2}(q^M) - c_i(q_{i1}^M + q_{i2}^M)$.*

In the equilibrium described in Proposition 1, all offers are accepted. When all offers are accepted, retailer j will buy exactly q_{ij}^M from manufacturer i . This is because once the contract is signed the marginal cost of acquiring brand i is zero up until $q_{ij} = q_{ij}^M$, while any quantity above this threshold leads to an infinitely high payment. This disciplines retailer j from selling more than q_{ij}^M of brand i , even if doing so would increase its flow profit (selling strictly less than q_{ij}^M will never be optimal).² When all contracts are accepted, the retailers will therefore sell the vector q^M on the downstream market and thereby maximize the industry profit in Stage 3.

To see that the contract offers and acceptance decisions specified in Proposition 1 constitute an equilibrium, consider first the acceptance decisions of retailer j . If it rejects one (or both) offers, the game ends and it earns a profit of zero, which is exactly what it gets in equilibrium. Therefore, neither retailer has a profitable deviation. Consider now a possible deviation from manufacturer h , where it offers a different set of contracts. If following the deviation all contracts are accepted, the rival manufacturer is assured its equilibrium profit, $\Pi_i(q^M)$. Since both retailers must earn non-negative profits in any continuation equilibrium, it follows that the profit of manufacturer h following a deviation is bounded from above by $\Pi^M - \Pi_i(q^M)$, which is exactly what it gets in equilibrium. If following the deviation from manufacturer h some contract offer is not accepted, the game ends and the profit of this manufacturer is zero. It follows that manufacturer h has no profitable deviation. We conclude that no firm has a profitable deviation.

As shown by Rey and Vergé (2010), if the manufacturers are restricted to offer two-part tariffs, industry profit will not be maximized in equilibrium. To maximize industry profit, wholesale prices must be above marginal cost in order to dampen downstream competition. However, when wholesale prices are at a level that maximizes industry profit, the manufacturers will have an incentive to free ride on each

² Quantity-rationing tariffs are used in exactly the same way in Section 4 of Miklós-Thal et al. (2011).

others upstream margins: Slightly reducing one's wholesale prices will only have a second order effect on industry profit, while having a first order negative effect on the rival manufacturer's profit. In equilibrium, wholesale prices will therefore be below the industry profit maximizing level.

The reason why a profit maximizing equilibrium can be supported by quantity rationing tariffs of the form studied in this article, is that the manufacturers get their entire profits through the fixed fees. Wholesale prices are equal to zero in the relevant quantity intervals, meaning that there are no upstream margins to free-ride on. A manufacturer will therefore have no incentive to deviate and induce more aggressive behavior by the retailers, even if this would increase the profit generated by the sale of its brand. Finally, the quantity ceilings ensure that a retailer will maximize the industry profit when competing downstream, even if behaving more aggressively would increase the profit generated by this retailer.

3.4 Discussion

When a retailer is able to sell imperfectly substitutable brands supplied by two independent manufacturers, it is typically able to extract rent from the manufacturers, even when the manufacturers offer contracts on a take-it-or-leave-it basis. The reason is that refusing to sell the brand of one manufacturer will tend to increase the flow profit the retailer can generate by selling the brand of the other manufacturer. This means that if the manufacturers try to extract the entire flow profit generated by a retailer, the retailer can accept only one contract and earn a strictly positive profit. Accepting only one contract is of course not an option in the intrinsic double common agency setting considered in the present article, and as a consequence the manufacturers are able to extract the entire industry profit. The market-break down assumption in the model in the previous section is however often unrealistic. Still, the outcome described in Proposition 1 would tend to hold also in other situations, as long as the manufacturers have access to equally efficient alternative retail channels at either location.

One could for example assume that a manufacturer first offered contracts to the common retailers, but offered a contract to an exclusive retailer at a given location if and only if the corresponding common retailer rejected the manufacturer's offer. Alternatively, one could imagine that a manufacturer had the opportunity to choose to sell directly to the consumers if a common retailer rejected its offer.

Both scenarios would make it less tempting for the common retailers to accept to trade with only one manufacturer, since the rejected manufacturer would have an incentive to compete aggressively (either directly or through its exclusive retailer) at the location in question. The reasoning behind Proposition 1 would then continue to hold: The relevant choice for a retailer would be to accept both or none of the contract offers, and the manufacturers could therefore extract the entire industry profit using quantity rationing tariffs.³

It should be noted that the tariffs studied in the present article do not contain any explicit restrictions on the prices charged (or quantities sold) by the retailers in the downstream market. They simply consists of a mapping from the quantity bought by a given retailer from a given manufacturer, to the payment made by the same retailer to the same manufacturer. In the setting considered in the current article, it is therefore little reason to believe that banning explicit contractual restrictions on the behavior of the retailers in the downstream market would lead to lower prices for consumers.

³ When the retailers can block the manufacturers' access to the downstream market, that is, when there are "retail bottlenecks", the analysis becomes significantly more complicated. Examining the extent to which quantity rationing tariffs can replicate the outcomes supported by RPM in such a setting is beyond the scope of this article.

Bibliography

- Marx, L. M. and G. Shaffer (2007). Upfront payments and exclusion in downstream markets. *RAND Journal of Economics* 38(3), pp. 823–843. DOI: 10.1111/j.0741-6261.2007.00114.x.
- Miklós-Thal, J., P. Rey, and T. Vergé (2011). Buyer power and intrabrand coordination. *Journal of the European Economic Association* 9(4), pp. 721–741. DOI: 10.1111/j.1542-4774.2011.01019.x.
- Rey, P. and T. Vergé (2010). Resale price maintenance and interlocking relationships. *The Journal of Industrial Economics* 58(4), pp. 928–961. DOI: 10.1111/j.1467-6451.2010.00439.x.
- Rey, P. and M. D. Whinston (2013). Does retailer power lead to exclusion? *RAND Journal of Economics* 44(1), pp. 75–81. DOI: 10.1111/1756-2171.12011.

Chapter 4

Menu contracts and the division of profits in vertical relationships

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Abstract

The article considers the use of menu contracts in vertical relationships. By offering publicly observable menus of contracts, competing retailers (manufacturers) can capture the entire industry profit when dealing with a common manufacturer (retailer).

4.1 Introduction

This article examines the effect of the ability to offer menu contracts on the division of profits in vertical relationships. Two industry structures are discussed. In the first, two competing retailers offer contracts to a common manufacturer. In the second, two competing manufacturers offer contracts to a common retailer. In both settings the ability to offer publicly observable menu contracts lets the firms with the bargaining power capture the entire maximal industry profit.

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Marx and Shaffer (2007) model an industry where two competing retailers offer contracts to a common manufacturer. They show that when contracts consists of single three-part tariffs, one of the retailers will be excluded in all equilibria.² Miklós-Thal et al. (2011) illustrate that when the contracts offered to the common manufacturer can be made contingent on exclusivity, there exists efficient equilibria in which both retailers are active. Rey and Whinston (2013) point out that as long as the retailers can offer menus of contracts for the manufacturer to choose from, there will always exists non-exclusionary equilibria in which each retailer gets its contribution to the industry profit, regardless of whether the contracts are allowed to be made explicitly contingent on exclusivity or not. The present article shows that when contract offers are public information, there also exist non-exclusionary equilibria in which each retailer gets strictly more than its contribution to the industry profit. In fact there exists equilibria in which the industry profit is maximized and the profit of the manufacturer is zero.

O'Brien and Shaffer (1997) and Bernheim and Whinston (1998) study a situation where two competing manufacturers offer contracts to a common retailer. They show that there always exists an equilibrium in which industry profit is maximized and each manufacturer gets its contribution to the joint profit, regardless of whether contract offers can be made contingent on exclusivity or not. The present article shows that when the manufacturers can offer publicly observable menus of contracts, there exists equilibria in which industry profit is maximized and entirely captured by the manufacturers, leaving the retailer with zero in profit. The contracts used to support such equilibria will be of the sort studied by Segal and Whinston (2003) and Martimort and Stole (2003), where it is the manufacturers that decide the quantities delivered to the retailer, after the retailer has chosen among the offered contracts.

The rest of the article is organized as follows. Section 4.2 analyzes the setting where two competing retailers offer contracts to a common manufacturer, while Section 4.3 analyzes the setting where two competing manufacturers offer contracts to a common retailer.

² In this setting a three-part tariff is a tariff that consists of an unconditional fixed fee that is paid regardless of whether the retailer decides to buy from the manufacturer or not, a conditional fixed fee that is paid only if the retailer buys a strictly positive quantity, and a per-unit wholesale price.

4.2 Competing retailers, common manufacturer

Let two competing retailers, R_1 and R_2 , buy and resell the product of a common manufacturer M . The retailers are assumed to be symmetric.³ The manufacturer incurs a cost $C(q_1, q_2) = c(q_1) + c(q_2)$, where q_i is the quantity sold to R_i , $i \in \{1, 2\}$. There are no costs at the retail level. The interaction between the firms is modeled as follows.

- (i) The retailers simultaneously offer menus of contracts to the manufacturer.
- (ii) The manufacturer chooses which contract, if any, to accept from each retailer. All contract offers and acceptance decisions are public information.
- (iii) Downstream competition takes place, and payments are made according to accepted contracts.

While general tariffs are allowed, special attention will be given to point tariffs, written (q_i, F_i) , which implies that a quantity q_i is delivered from M to R_i in exchange for a transfer F_i . The equilibrium concept used in this article will be that of subgame-perfect Nash equilibrium (hereafter equilibrium).

It will be convenient to follow section 4 of Rey and Whinston (2013) and let $\rho(q_i, q_{-i})$ denote the revenue generated by R_i 's sale. Assume that $\rho(q_i, q_{-i})$ is always non-increasing in q_{-i} . The joint profit of the three firms is then $\Pi(q_1, q_2) \equiv \sum_i \rho(q_i, q_{-i}) - c(q_1) - c(q_2)$. Let the maximal joint profit be denoted Π^* , and suppose that it is obtained when $(q_1, q_2) = (q^*, q^*)$. If only R_i buys from M , the joint profit is $\Pi(q_i, 0) \equiv \rho(q_i, 0) - c(q_i)$. Assume this exclusivity profit is maximized at $q_i = q^m$, where it attains the value Π^m .

It is assumed that the the retailers are substitutes in creating profit:

Assumption 1. $2\Pi^m > \Pi^* \geq \Pi^m$.

The (marginal) contribution of each retailer to the industry profit is given by $\Pi^* - \Pi^m \geq 0$. To see how the retailers can use menus of contracts to each extract more than this contribution, let retailer R_i ($i = 1, 2$) offer the menu $C_i = (C_i^C, C_i^E)$ to M , where M freely can choose which (if any) of the two contracts offered by R_i to accept.

³ This assumption is made for expositional simplicity. The results of this section would hold also with asymmetric retailers.

In these menus C_i^C consists of a point tariff $(q^*, c(q^*))$, which implies that R_i gets q^* and pays the production cost $c(q^*)$. C_i^E on the other hand, gives R_i the choice between $(0, 0)$, that is buying nothing and paying nothing, and $(q^m, \rho(q^m, 0))$. The latter option implies that R_i gets the quantity that is optimal under exclusivity and pays the corresponding revenue that would arise under exclusivity. If M accepts C_1^C and C_2^C , both retailers will buy from M in stage (iii), and the industry profit will be Π^* . Since M in this case only gets transfers equal to the cost of production, it will get zero in profit. R_i will get a profit of $\rho(q^*, q^*) - c(q^*) = \frac{1}{2}\Pi^* > \Pi^* - \Pi^m$.⁴

I claim that there indeed exists an equilibrium in which the retailers offer the menus $C_1 = (C_1^C, C_1^E)$ and $C_2 = (C_2^C, C_2^E)$ and in which M accepts C_1^C and C_2^C . To see that this is the case, note that if M accepts C_i^E from R_i and rejects the offer from R_{-i} , R_i will in stage (iii) be indifferent between buying q^m from M (and paying the fee $\rho(q^m, 0)$) and buying nothing and paying nothing. Now, let the strategy of R_i specify the following play in stage (iii). If the retailers offer $C_1 = (C_1^C, C_1^E)$ and $C_2 = (C_2^C, C_2^E)$ and M accepts C_i^E , R_i will choose not to buy (choose the point tariff $(0, 0)$) in stage (iii). This is credible, since the most it could get by choosing $(q^m, \rho(q^m, 0))$ is zero. If however the rival retailer offers any other menu of contracts in stage (i), R_i 's strategy specifies that it will choose to buy q^m from M in stage (iii) (and pay the entire ensuing revenue $\rho(q^m, 0)$) if C_i^E is accepted and the rival's offer is rejected. This is also credible, since instead choosing the tariff $(0, 0)$ would also give zero in profit.

Now, observe that M cannot do better than to accept C_1^C and C_2^C . The strategy of R_i implies that it will never buy using the point tariff $(q^m, \rho(q^m, 0))$, so if R_i buys from M in stage (iii), it must be using the tariff $(q^*, c(q^*))$. But when both R_i ($i = 1, 2$) either are inactive or chooses $(q^*, c(q^*))$, the profit of M is zero, implying that M has no profitable deviation.

Further, R_i has no profitable deviation in stage (i) given the strategy of the rival. By offering a different contract (whether a menu of point tariffs or any other type of tariff) it enables M to accept only C_{-i}^E and earn Π^m , since following such a deviation R_{-i} would choose $(q^m, \rho(q^m, 0))$ over $(0, 0)$. This means that following any deviation in stage (i) the profit of R_i is bounded from above by $\Pi^* - \Pi^m$, which is strictly below what it gets in the proposed equilibrium. The following proposition sums up.

⁴ The equality follows from the assumed symmetry of the retailers, while the inequality follows from Assumption 1.

Proposition 1. *When two competing retailers can offer menus of contracts to a common manufacturer, there exists an equilibrium in which industry profit is maximized and the profit of the manufacturer is zero.*

In the equilibrium specified above, any deviation from R_i is preempted, since the strategy of the rival retailer entails a credible threat: Following a deviation from R_i in stage (i), the rival retailer will accept to buy q^m from M and leave M the entire ensuing profit. This makes it impossible for R_i to profitably deviate in a way that increases the joint profit of itself and M , even though their joint equilibrium profit of $\frac{1}{2}\Pi^*$ is strictly below what they potentially could obtain through an exclusive relationship, namely Π^m .⁵

Corollary 2 in Rey and Whinston (2013) states that, given that three-part tariffs are part of the allowable set of contracts, a retailer can never get more than its contribution to the industry profit in any equilibrium. Proposition 1 shows that this is not the case, as long as all contract offers are public information.⁶ In fact, the retailers can always extract the entire industry profit, regardless of the size of their individual contributions to the industry profit.

4.3 Competing manufacturers, common retailer

We have seen how menus of contracts can be used by competing retailers to extract the maximal industry profit when offering contracts to a common manufacturer. One may wonder whether similar strategies can be used by competing manufacturers to extract the entire industry profit if they are able to offer public menus of contracts to a common retailer. The answer is yes, given that we allow the manufacturers to choose the quantities to be delivered after the common retailer has chosen which contracts to accept (in the spirit of Segal and Whinston (2003) and Martimort and Stole (2003)).

⁵ By making the choice of continuation equilibrium dependent on history of play, the strategies above resemble strategies that support cooperative outcomes in repeated oligopoly games.

⁶ The result in Proposition 1 relies on the assumption that also rejected contract offers are public information. If only accepted contracts become public information, the profit the manufacturer can get by only accepting a contract from one retailer will be independent of the offer from the rival retailer. As a consequence, the equilibrium profit of each retailer will be restricted to its contribution to the industry profit. While Marx and Shaffer (2007) and Miklós-Thal et al. (2011) assume that all offers are public information, Rey and Whinston (2013) only explicitly state that accepted contracts become public information. It should be noted that if the manufacturer could choose which (if any) contract offers to make public, Proposition 1 would continue to hold.

To see this, assume that two symmetric manufacturers, M_1 and M_2 , offer contracts to a common retailer R .⁷ Let $\rho(q_i, q_{-i})$ be the revenue generated by the sale of the product of M_i , which has a production cost of $c(q_i)$. Retail cost is again assumed to be zero. As before, denote the maximal joint profit in the industry by Π^* , and suppose that it is obtained when $(q_1, q_2) = (q^*, q^*)$. Assume that industry profit when only M_i is active is maximized when $q_i = q^m$, where it attains the value Π^m . The manufacturers are assumed to be substitutes in creating profits, implying that $2\Pi^m > \Pi^* \geq \Pi^m$.

The interaction between the firms is modeled as follows.

- (i) The manufacturers simultaneously offer menus of contracts to the retailer.
- (ii) The retailer chooses which contract, if any, to accept from each manufacturer. All contract offers and acceptance decisions are public information.
- (iii) Manufacturers with accepted contracts choose the quantity to be delivered to the retailer, and payments are made according to accepted contracts.

To see that there exists an equilibrium in which industry profit is maximized and the retailer gets zero in profit, let the M_i offer a menu $C_i = (C_i^C, C_i^E)$ for R to choose from. If accepted, C_i^C specifies an upfront payment equal to $\rho(q^*, q^*) - c(q^*)$, paid by R to M_i independently of the quantity delivered in stage (iii). In addition C_i^C gives M_i the choice in stage (iii) between the point tariff $(q^*, c(q^*))$, that is delivering q^* in exchange for a transfer equal to the production cost $c(q^*)$, and $(0, 0)$, that is not delivering anything and not getting a transfer. C_i^E does not include an upfront payment and will, if accepted, give M_i the choice between $(0, 0)$ and $(q^m, c(q^m))$ in stage (iii), where the latter consists of M_i delivering the quantity optimal under exclusivity, q^m , and getting a transfer equal to the cost of production, $c(q^m)$.

Let the strategy of M_i specify the following play in stage (iii). If the menus $C_1 = (C_1^C, C_1^E)$ and $C_2 = (C_2^C, C_2^E)$ are offered, M_i will refuse to deliver anything, that is choose $(0, 0)$, unless C_1^C and C_2^C are accepted, in which case it will choose $(q^*, c(q^*))$. This is credible, since M_i is always indifferent between choosing $(0, 0)$ and the alternative point tariff. Further, the strategy specifies that if the rival M_{-i} offers any other menu of contracts, M_i will choose $(q^m, c(q^m))$ if C_i^E is accepted.

I claim that there exists an equilibrium in which $C_1 = (C_1^C, C_1^E)$ and $C_2 = (C_2^C, C_2^E)$ are offered and C_1^C and C_2^C are accepted. To see this, note that given

⁷ Again, the results of this section would hold also with asymmetric manufacturers.

the manufacturers' strategies, R can do no better than to accept C_1^C and C_2^C . This gives each manufacturer a profit (through the upfront payment) of $\rho(q^*, q^*) - c(q^*) = \frac{1}{2}\Pi^* > \Pi^* - \Pi^m$ and R a profit of zero. Accepting any other combination of contracts would lead to both manufacturers delivering a quantity equal to zero, in which case the profit of R is bounded from below at zero.

Further, M_i has no profitable deviation. By deviating in stage (i) and offering a different menu of contracts, it would enable R to accept C_{-i}^E and get a profit of Π^m , since the rival M_{-i} would choose $(q^m, c(q^m))$ over $(0, 0)$ following such a deviation. It follows that the profit of M_i following a deviation is bounded from above by $\Pi^* - \Pi^m$, which is strictly below what it gets in the proposed equilibrium.

We get the following proposition.⁸

Proposition 2. *When two competing manufacturers can offer menus of contracts to a common retailer, and the manufacturers decide the quantities to be delivered in the final stage of the game, there exists an equilibrium in which industry profit is maximized and the profit of the retailer is zero.*

⁸ As was the case in the previous section, this result would continue to hold if the retailer could choose which contract offers to make public.

Bibliography

- Bernheim, B. D. and M. D. Whinston (1998). Exclusive Dealing. *Journal of Political Economy* 106(1), pp. 64–103. DOI: 10.1086/250003.
- Martimort, D. and L. Stole (2003). Contractual externalities and common agency equilibria. *Advances in Theoretical Economics* 3(1). DOI: 10.2202/1534-5963.1037.
- Marx, L. M. and G. Shaffer (2007). Upfront payments and exclusion in downstream markets. *RAND Journal of Economics* 38(3), pp. 823–843. DOI: 10.1111/j.0741-6261.2007.00114.x.
- Miklós-Thal, J., P. Rey, and T. Vergé (2011). Buyer power and intrabrand coordination. *Journal of the European Economic Association* 9(4), pp. 721–741. DOI: 10.1111/j.1542-4774.2011.01019.x.
- O’Brien, D. P. and G. Shaffer (1997). Nonlinear supply contracts, exclusive dealing, and equilibrium market foreclosure. *Journal of Economics & Management Strategy* 6(4), pp. 755–785. DOI: 10.1111/j.1430-9134.1997.00755.x.
- Rey, P. and M. D. Whinston (2013). Does retailer power lead to exclusion? *RAND Journal of Economics* 44(1), pp. 75–81. DOI: 10.1111/1756-2171.12011.
- Segal, I. and M. D. Whinston (2003). Robust predictions for bilateral contracting with externalities. *Econometrica* 71(3), pp. 757–791. DOI: 10.1111/1468-0262.00425.